



*transmission media is shared between multiple users. or multiple signals from the same user but from diff App that goes to same transmission media.

Chapter 6

Bandwidth Utilization: Multiplexing and Spreading

channel has BW
so if channel is BW is 100
for signals that are 100
I cannot transfer more than one
user.

transmission media
لا يمكن أن يكون
العرض النطاقي

BW for users that share
transmission media

I have to share [utilize]
between different users.

Note

share

Bandwidth utilization is the wise use of available bandwidth to **achieve specific goals.**

Efficiency can be achieved by

multiplexing

Privacy and anti-jamming can be

achieved by **spreading**.

each is assigned different carrier frequency.
same data pattern sent on
different frequency.

[same user do spreading
on all BW]

user share BW between multiple users

share BW between multiple users
مستخدم واحد يستخدم BW مشترك

[multiple users use BW]
المستخدمين يستخدمون BW مشترك

one signal send
all over the BW

according to the
goal
وفقاً للهدف

BW utilization

multiple output

single output

selection-lines

A number line from 0 to 12. Points 1, 2, and 3 are marked. Point 1 is at 0, point 2 is at 8, and point 3 is at 12. The segments are labeled 4, 8, and 12.

- * كل User يملك
مقدار ال
التي يملكه فقط

- sharing frequently between users

- sharing media based on time.
 ① user specifies user ②
 link or data bit rate. you are using all the
 1000 b/s. Data is 1000 b/s. high efficiency and
 performance

وال network بعت 100bps وال الينته 5sec في 2. 1sec. تبقيها 4/4.

Multiplexing

- If the link bandwidth is greater than the bandwidth needed by a single device, then the link can be shared by multiple devices
- Multiplexing is the set of techniques used to simultaneously transmit multiple signals across a single data link
- Multiplexing creates multiple transmission channels over a single communication link
- Several recent transmission media such as optical fibers and satellite microwave links have much higher bandwidth than the average bandwidth needed for most applications
- It is essential to optimize the utilization of a given bandwidth to reduce costs and prevent wastage. That is, to improve the cost-effectiveness and utilization of the resources

Goal of optimization :-

①

②

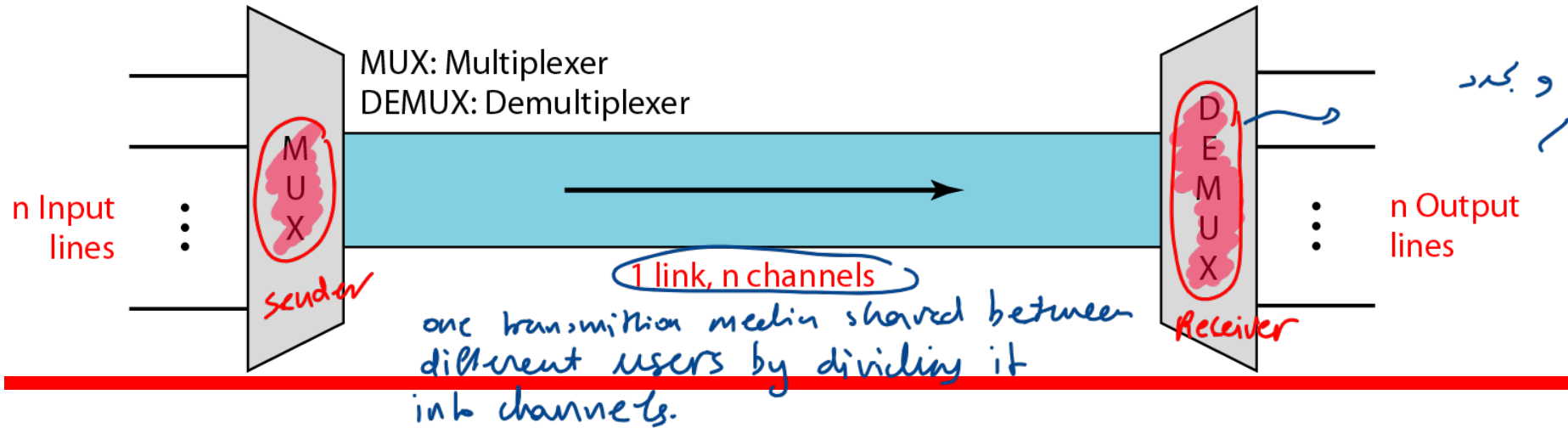
BW has Range. & number of users is increasing

6.4

cannot assign more → limited by all users need

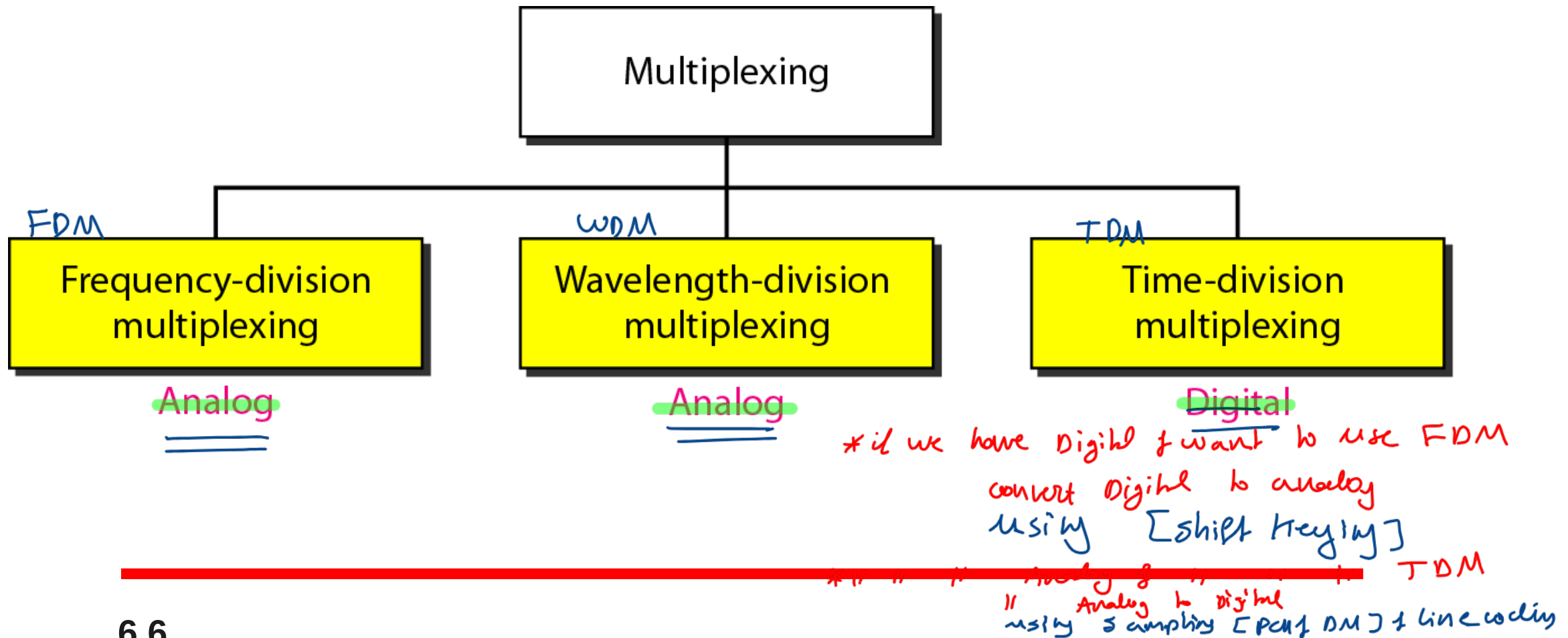
data is sent over channel user is not aware spectrum limited all users need

250kHz primary user
 BW between different users
 sharing 1 secondary + primary
 called cognitive Radio networks.
 2 subcarrier freq
 100kHz secondary user
 Figure 6.1 Dividing a link into channels



- The multiplexer (MUX) combines several channels into one stream (many-to-one conversion)
- The demultiplexer (DEMUX) separates the stream back into the original channels (one-to-many conversion)
- The **link** refers to the **physical medium**
- The **channel** refers to the **portion of the link** that carries the transmission between any pair of lines {have carrier frequency 150} * n users, so n channels each channel to one user.

Figure 6.2 *Categories of multiplexing*



Frequency Division Multiplexing (FDM)

- FDM is an analog technique that can be applied when the bandwidth of the link in Hertz ^{transmission media} is ^{or equal} greater ^{sum} than the combined bandwidths of the individual signals to be transmitted.
 10 signals each one 4K, the BW at least should be 40K.
- Each signal is modulated with a different carrier frequency ^{per user} FSK
- The modulated signals are then combined to form a composite signal, which is transmitted over the link
 analog → Digital → Analog
 FDM
- The carrier frequencies should be separated enough using guard bands to
 prevent overlapping or interfering with neighboring signals

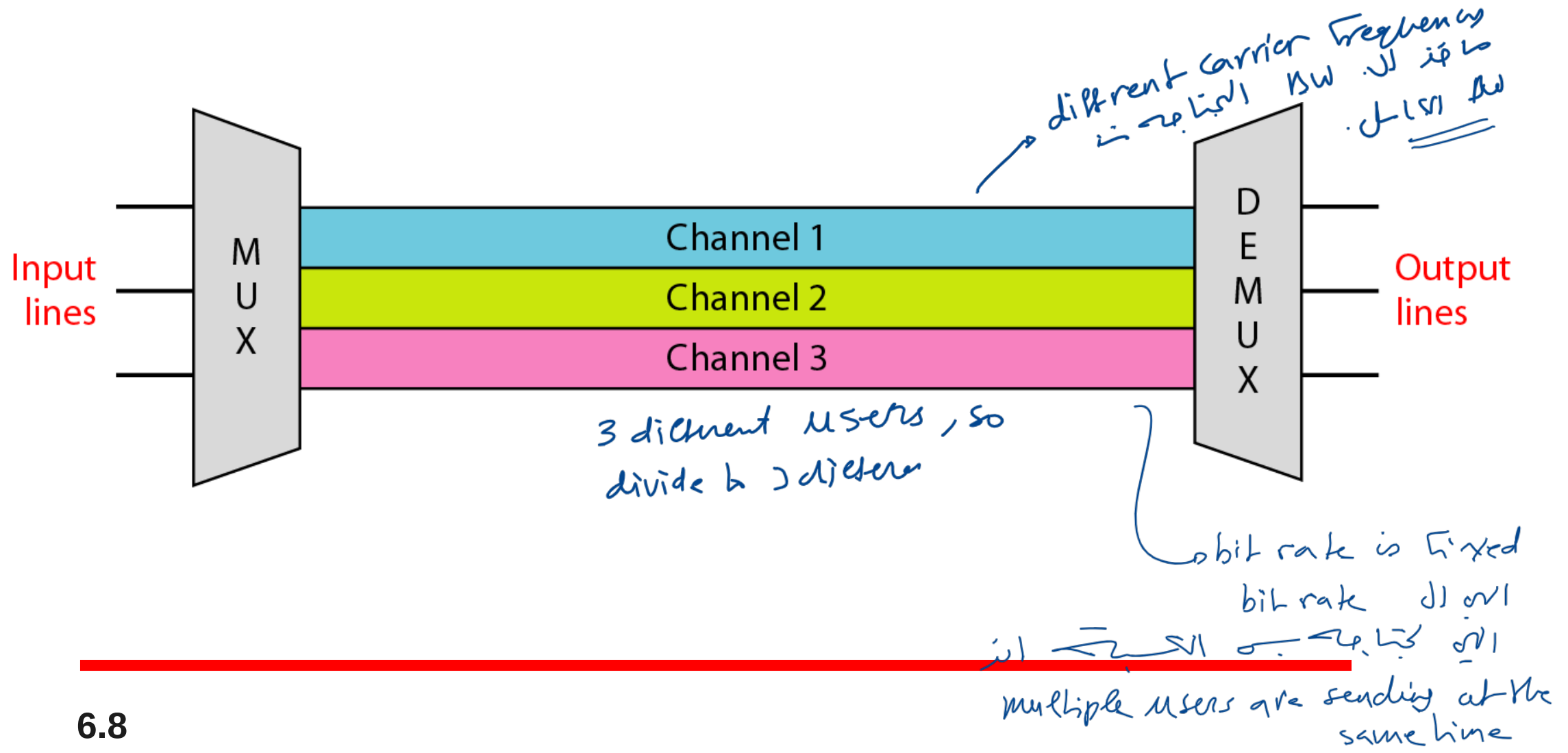
6.7 $(9 \times 4) + 40 = 76 \text{ kHz}$
 I will use

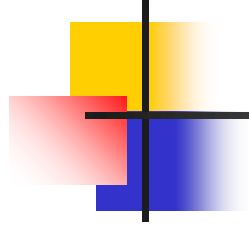
10 users → 4 kHz
I have 100 kHz, I will use 40 kHz for users
4 kHz to give user & user to give gap between

signal line
different signals
different users
or
different carrier freq.

$$\text{BW utilization} = \frac{76}{100} = 76\%$$

Figure 6.3 *Frequency-division multiplexing*

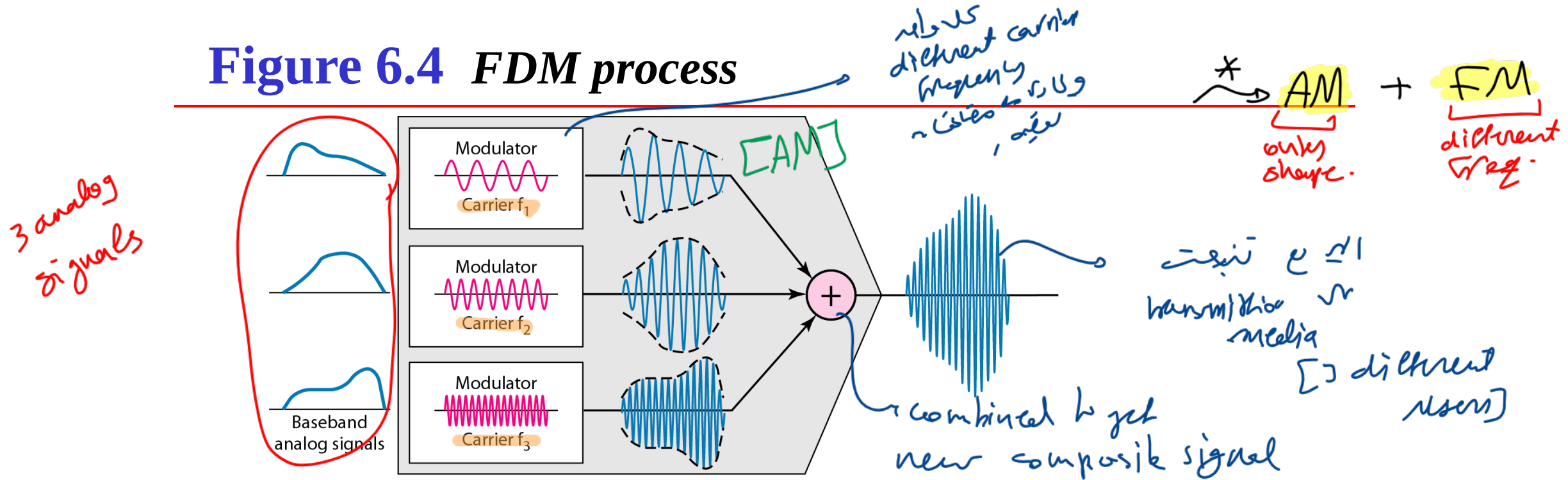




Note

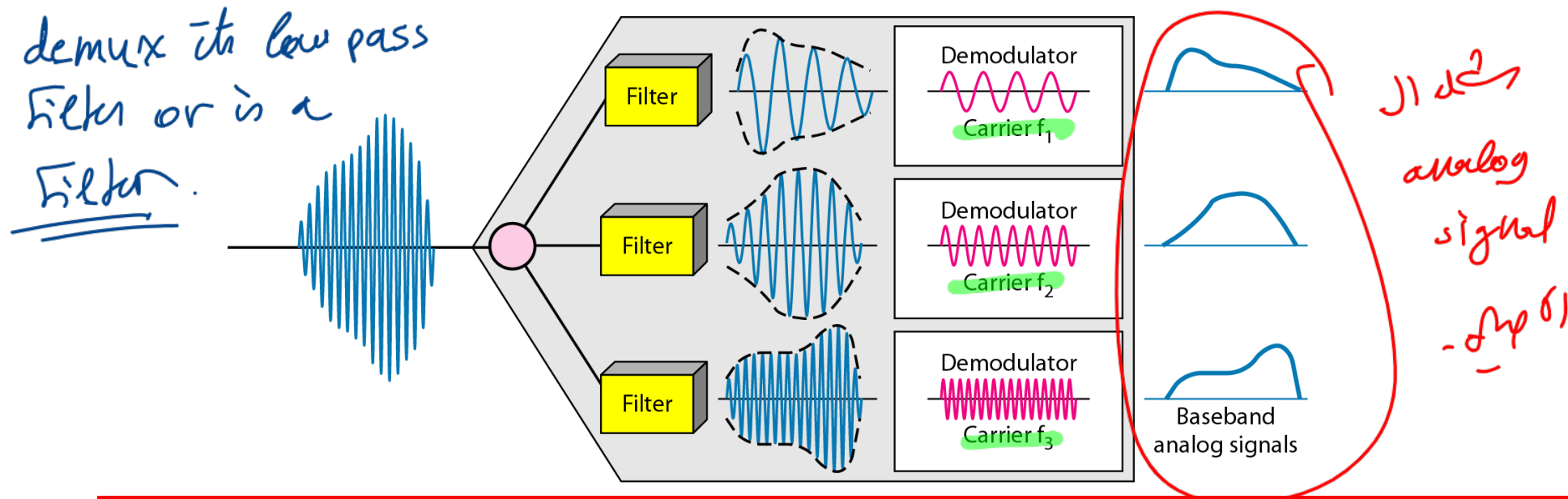
FDM is an analog multiplexing technique that combines analog signals.

Figure 6.4 FDM process



- The input signals have similar frequency ranges and bandwidth requirements
- Each signal is shifted in frequency away from the other signals
- The modulated input signals are then combined to form a single composite signal, whose bandwidth is equal to, **at least**, the sum of the all input signals' bandwidths

Figure 6.5 *FDM demultiplexing example*



- A series of filters are used to decompose the multiplexed signal into its originally modulated signals
- The individual modulated signals are demodulated and shifted back to their original frequency ranges
- The original signals are then passed to their intended recipients

Example 6.1

- Assume that a ^{user}voice channel occupies a bandwidth of 4 kHz.
- We need to combine three voice channels into a link with a bandwidth of 12 kHz, from 20 to 32 kHz.
- Show the configuration, using the frequency domain. Assume there are no guard bands.

**can we use FDM?*

$$\begin{array}{r} 4+4+4 \\ 12 \end{array} = \begin{array}{r} 12 \\ 12 \end{array}$$

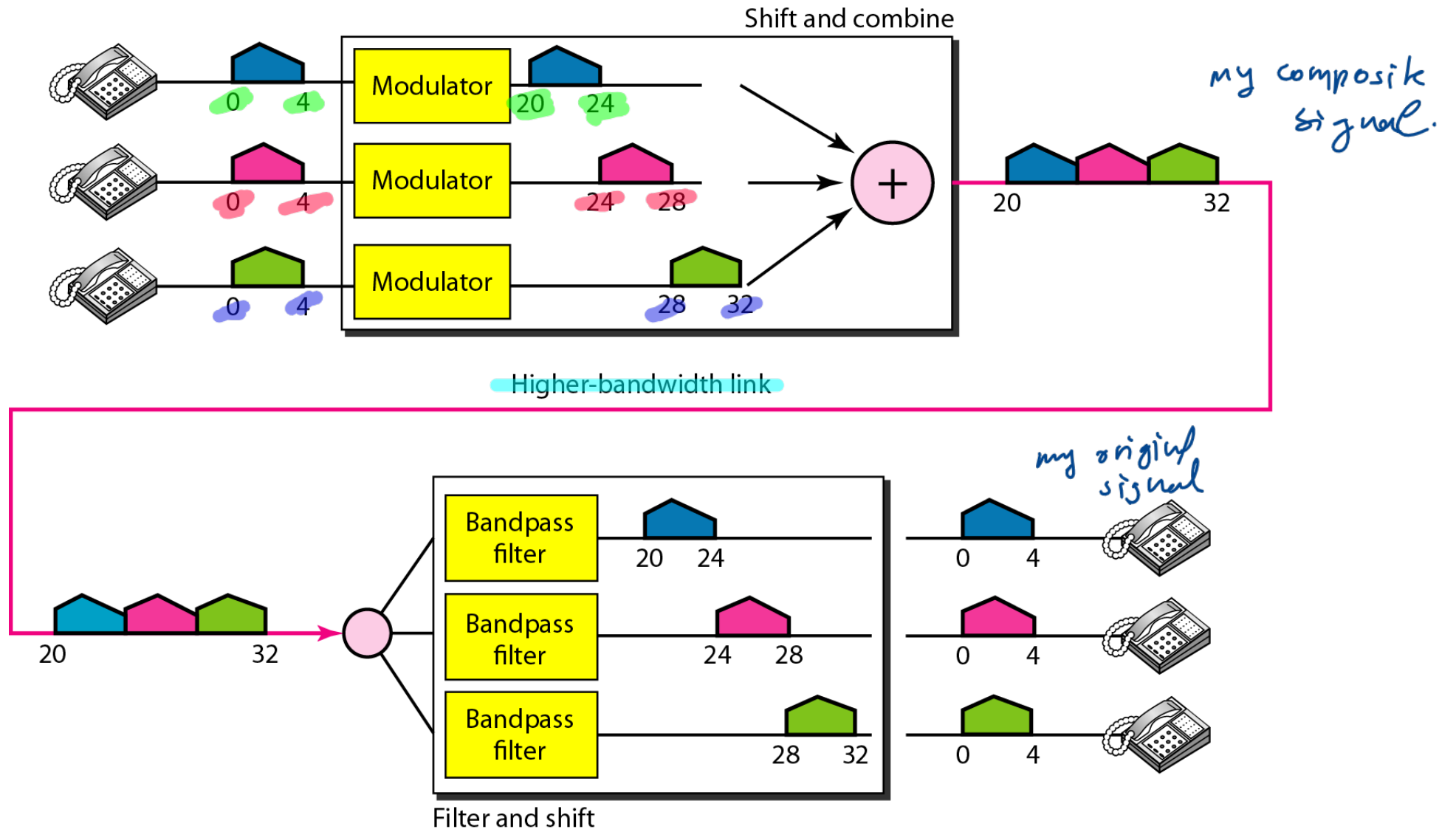
✓ yes, but without guard bands!

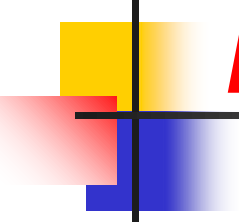
Solution

- We shift (modulate) each of the three voice channels to a different bandwidth, as shown in Figure 6.6. *[needs three channels with three carrier frequency]*
- We use the ~~20- to 24-kHz bandwidth for the first channel~~, the ~~24- to 28-kHz bandwidth for the second channel~~, and the ~~28- to 32-kHz bandwidth for the third one~~.
- Then we combine them as shown in Figure 6.6.

$$\begin{array}{l} f_{c1} = 22 \\ f_{c2} = 24 \\ f_{c3} = 26 \end{array}$$

Figure 6.6 Example 6.1





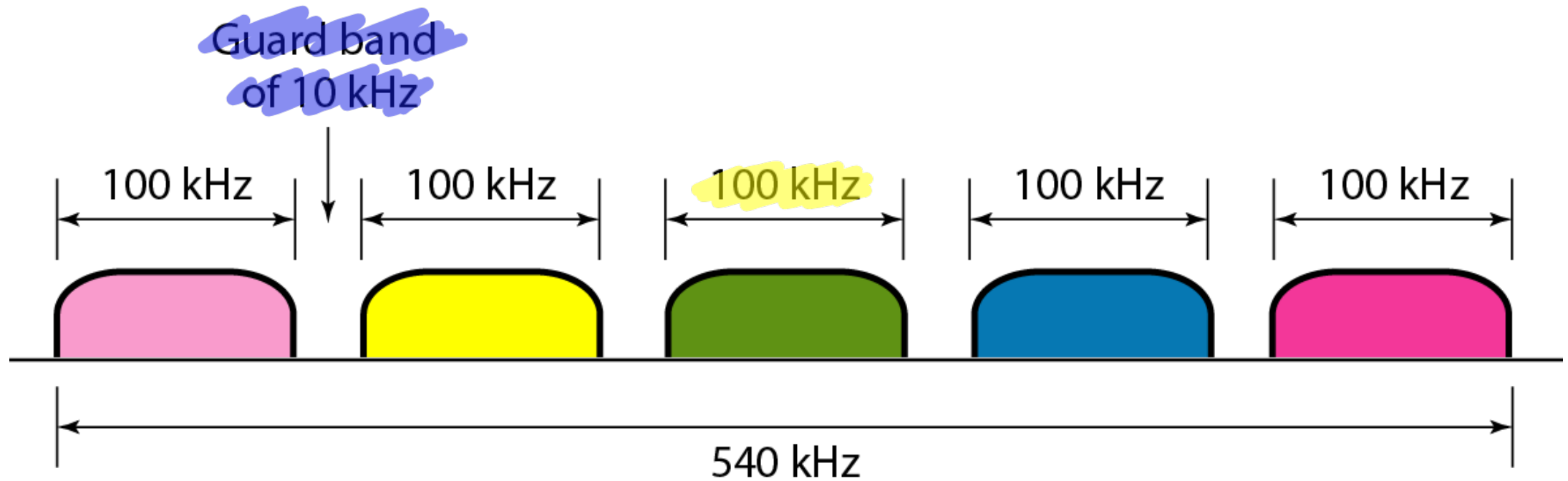
Example 6.2

- Five channels, each with a 100-kHz bandwidth, are to be multiplexed together.
- What is the minimum bandwidth of the link if there is a need for a guard band of 10 kHz between the channels to prevent interference?

Solution

- For five channels, we need at least four guard bands.
- This means that the required bandwidth is at least
$$5 \times 100 + 4 \times 10 = 540 \text{ kHz},$$
- as shown in Figure 6.7.

Figure 6.7 *Example 6.2*



Example 6.3

- Four data channels (digital), each transmitting at 1 Mbps, use a satellite channel of 1 MHz. analog signal
 - { Design an appropriate configuration, using FDM. }
- $\# \text{ of users} = 4$
 $\text{BW for each user} = 250 \text{ kHz}$
 $\text{Available BW} = 1 \text{ MHz}$

Solution

- The satellite channel is analog. 250kHz ہر ایک کے واسطے ہے۔ between different users کے ساتھ۔
- We divide it into four channels, each channel having a 250-kHz bandwidth. کیسے؟ ہر ایک کے واسطے ہے۔ Digital اور analog - کیسے؟
- Each digital channel of 1 Mbps is modulated such that each 4 bits are modulated to 1 Hz. 1MHz give 250kHz.
- One solution is to use 16-QAM modulation. B_{min} = 250 KHz
- Figure 6.8 shows one possible configuration. digital to analog → ASK / FSK / PSK / QAM.

ASK → $B_{\min} = \frac{N}{r} = \frac{1M}{4} = 250K$ → MASK where $r=4$.

FSK → No sol

16-QAM → with $r=4$

6.16

* PSK → MASK where $r=4$

Figure 6.8 Example 6.3

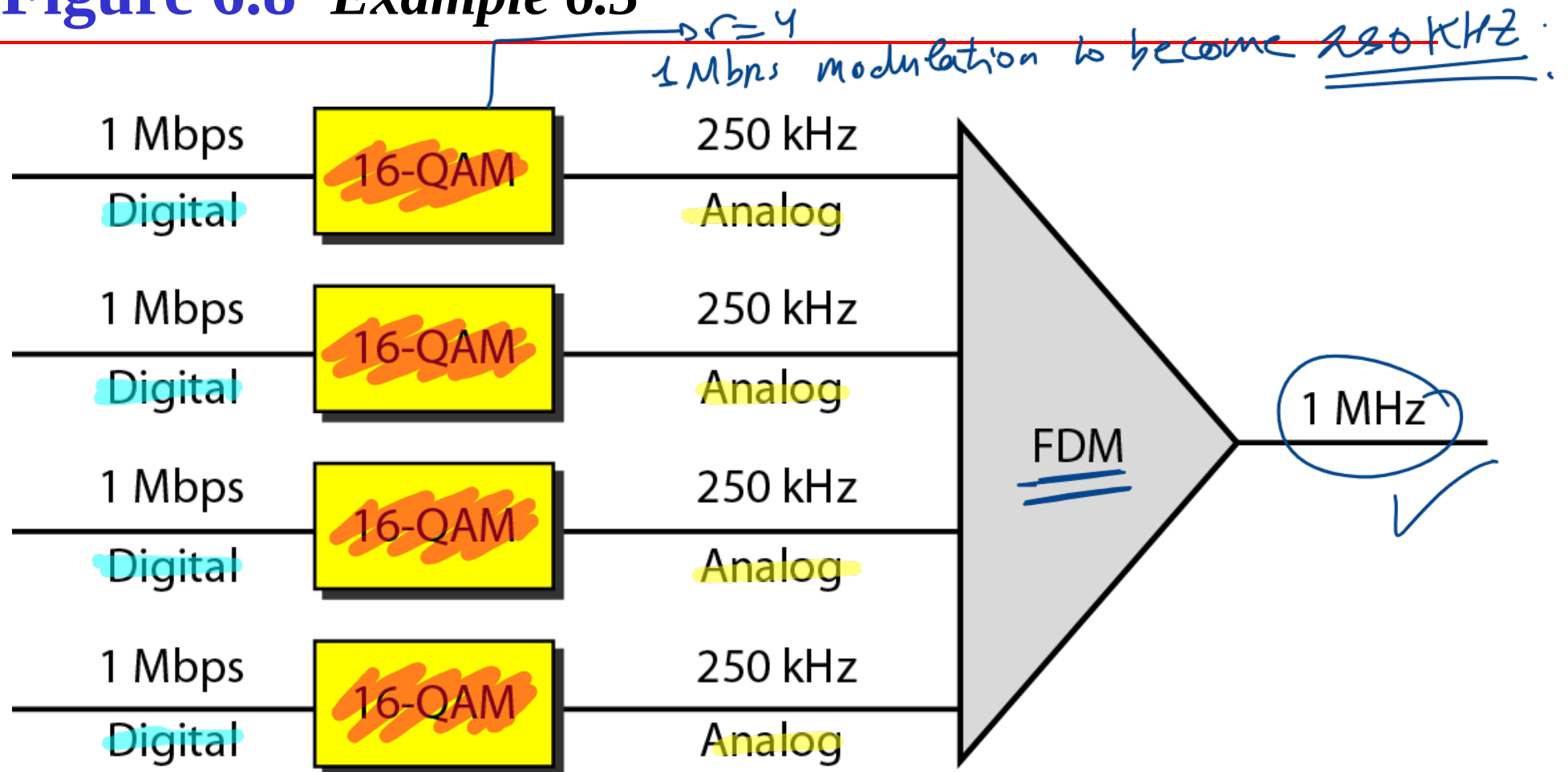
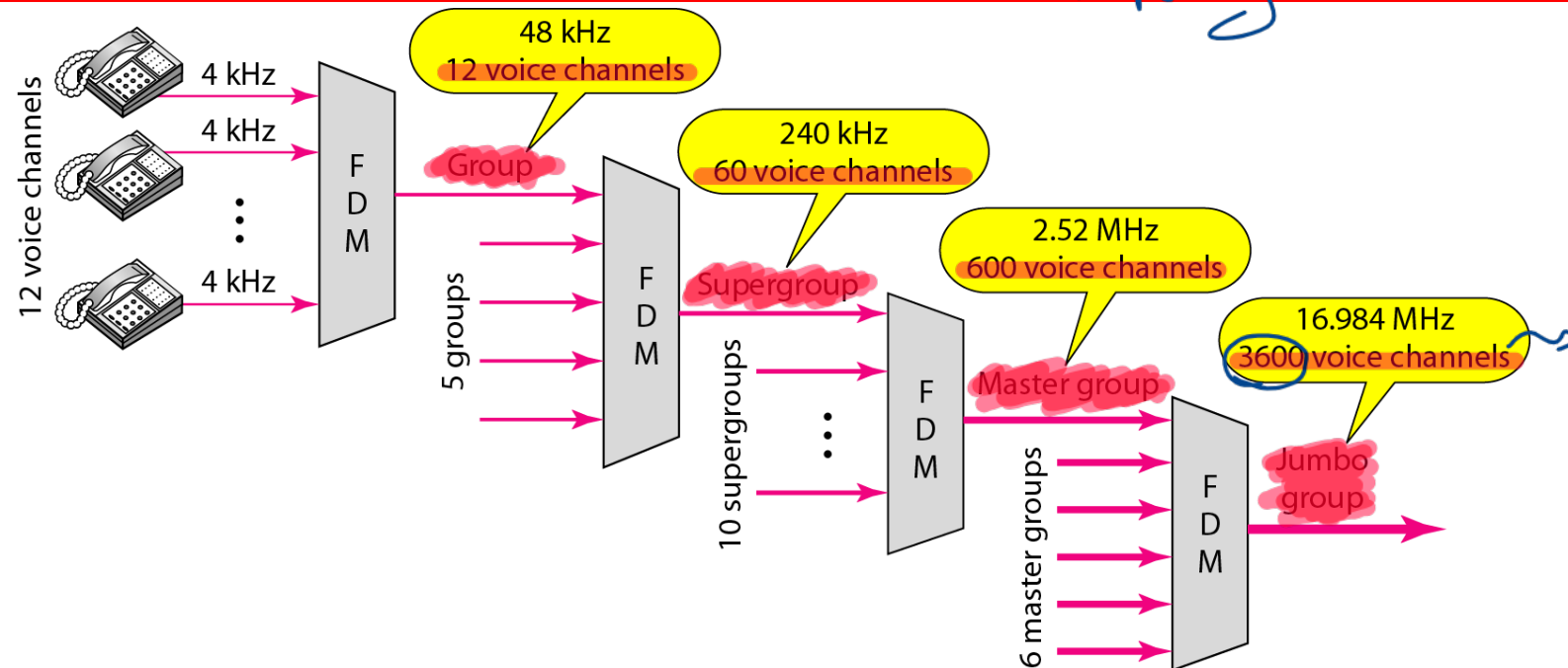


Figure 6.9 Analog hierarchy *to do frequency division multiplexing.*



- Used by telcos to optimize the efficiency of the infrastructure
- Combines multiple low bandwidth lines into higher bandwidth
- Fewer but larger lines are used:
 - **Advantages:** optimized installation cost and easier long haul management
 - **Disadvantages:** less diversity and more down lines in case of a failure

Other Applications of FDM

- **AM and FM radio broadcasting:** *different users in. frequency* *تفاوت در فرکانس*
 - Air is the transmission medium
 - A limited bandwidth is assigned for each modulation technique
 - Each station shifts its signal into a predefined band
 - Guard bands are automatically predefined
 - All signals are transmitted over the air simultaneously as if they are a single composite signal
 - The radio receiver filters out the composite signal to a single desired modulated signal, which is demodulated
- **TV broadcasting:** similar to AM and FM in concept *تفاوت در فرکانس*
- First generation of **cellular telephones (called AMPS):**
 - **3 kHz** voice signal is FM modulated
 - Two 30 kHz channels are assigned for each user to receive and transmit
 - Carrier frequencies are dynamically assigned/reassigned based on usage

Example 6.4

- The Advanced Mobile Phone System (AMPS) uses two bands.
- The first band of 824 to 849 MHz is used for sending, and 869 to 894 MHz is used for receiving.
- Each user has a bandwidth of 30 kHz in each direction.
- How many people can use their cellular phones simultaneously?

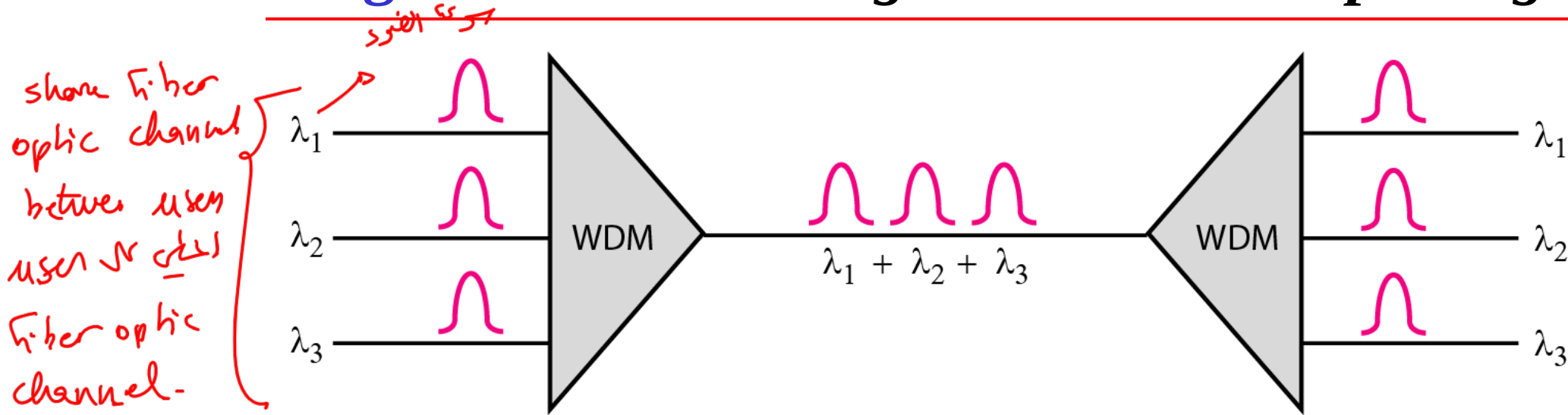
Solution

- Each band is 25 MHz ($849 - 824 = 25$ & $894 - 869 = 25$)
- If we divide 25 MHz by 30 kHz, we get 833.33 sub channels means 833 user at the same time
- In reality, the band is divided into 832 channels
- Of these, 42 channels are used for control, which means only 790 channels are available for cellular phone users

* FDM works on Analog
* if we have digital do modulation first

790 user at the same is calling

Figure 6.10 Wavelength-division multiplexing (WDM)

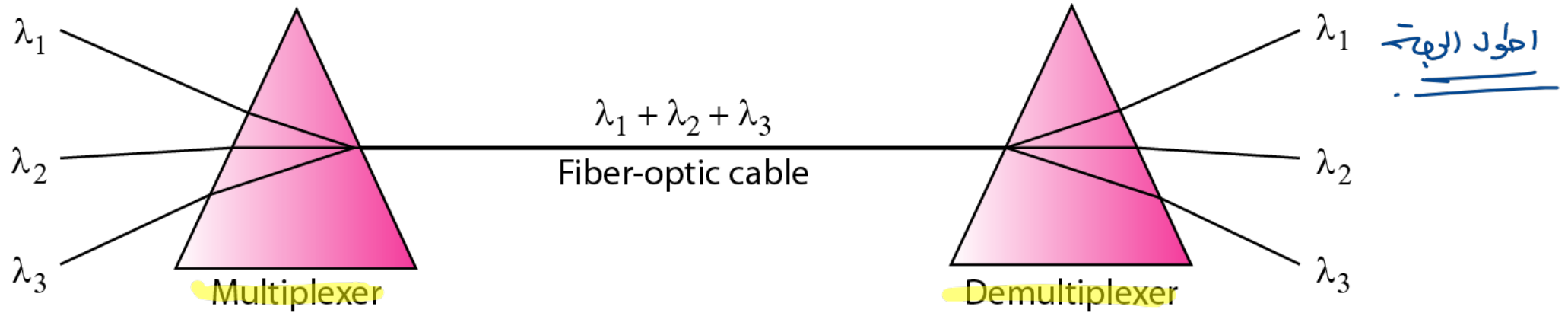


*Adv for FDM:- multiple users sending at the same time, each on different freq. so increasing the efficiency

Adv for FDM:- BW b/w is fixed $\Rightarrow$ rate fixed / speed constant.

- Designed for the high-bandwidth fiber-optic cable
 - Similar in concept to FDM except that it is used to mux and demux optical signals transmitted over fiber-optic channels
 - Optical signals are very high frequency signals that are usually defined by their wavelengths
- sharing link at different wavelengths
الحال موجه مختلف

Figure 6.11 *Prisms in wavelength-division multiplexing and demultiplexing*



-
- Multiple narrow band light signals are combined into a single wide band light signal using a prism
 - The prism bends a beam of light based on the signal frequency and the angle of incidence
 - The prism is used both as mux and demux

Note

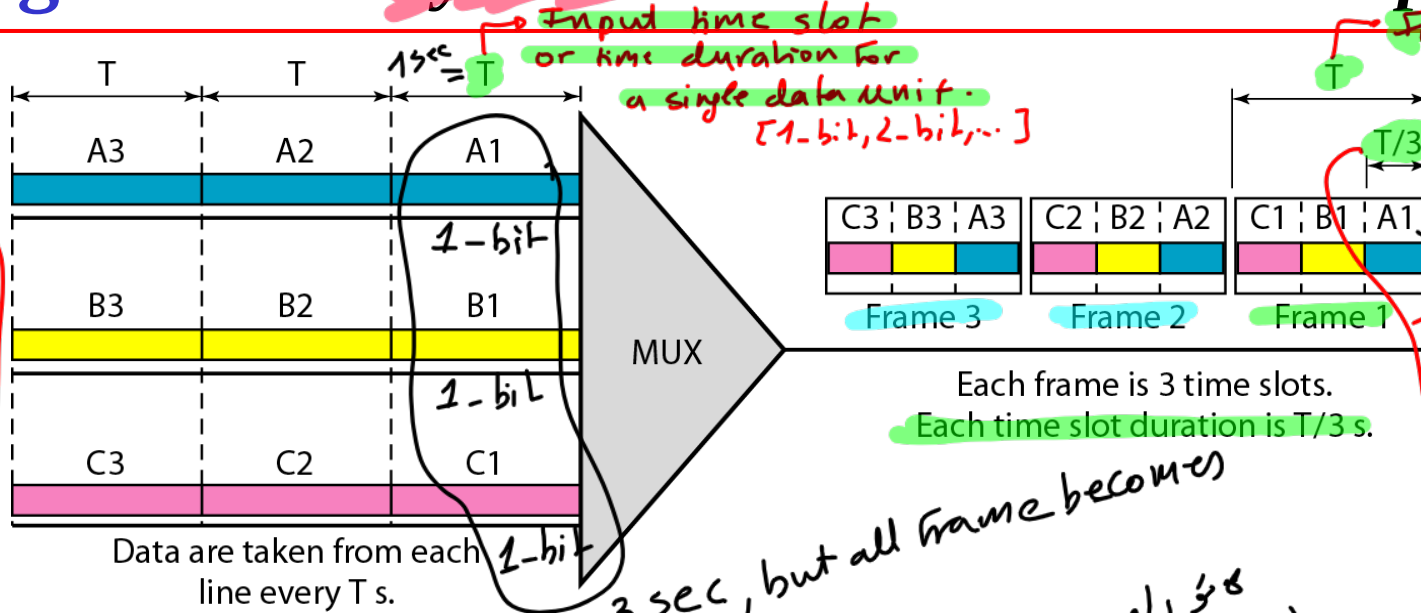
TDM is a digital multiplexing technique for combining several low-rate channels into one high-rate one.

Figure 6.13 Synchronous time-division multiplexing

*Three users
A, B & C

output bit rate
at least 30mbps

10M
10M
10M



3 users, with same bit rate always have data to send. The user always have slot either it sends or not.

Input time slot or time duration for a single data unit. [1-bit, 2-bit, ...]

Frame time T

$3 \times \frac{n}{n}$ data unit from each user.

size of the frame is n [number of users]

3 data units
3 bits or
3 bytes

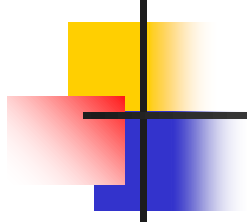
Each frame is 3 time slots.
Each time slot duration is $T/3$ s.

3 sec, but all frame becomes one sec.

- The data flow of each connection is divided into units
- Data units are combined into frames
- The frame consists of, at least, one data unit from each connection
- For n connections, the frame has at least n time slots
- The data rate of the link is n times faster than each connection and hence the unit duration is n times shorter

output = 1 sec
Input = 1 sec
output = $\frac{1}{\text{# of users}}$ or $\frac{1}{\text{Frame size}}$
or $\frac{T}{\text{number of users}}$

output bit rate = $\sum$ bit rate for each user.



frame rate = # of frames per sec

3 users $\rightarrow$ 3 frames

= users 110 Mbps

= 10 Mbps

= $\frac{1}{\text{frame duration}}$ better

Note

users with similar data rates.

In synchronous TDM, the data rate of the link is n times faster, and the unit duration is n times shorter.

output bit rate = $\sum$ bit rate of the input

$T \& T$
3

Example 6.5

- In Figure 6.13, the data rate for each input connection is 1kbps.
- If 1 bit at a time is multiplexed (a unit is 1 bit), what is the duration of (a) each input slot, (b) each output slot, and (c) each frame?

Solution

- We can answer the questions as follows:
 - a. The data rate of each input connection is 1 kbps. This means that the bit duration is $1/1000$ s or 1 ms $\rightarrow$ input time slot is 1 ms. *each data unit has duration of 1ms*
 - b. The duration of each output time slot is $1/3$ of the input time slot = $1/3$ ms.
 - c. Each frame carries three output time slots. So the duration of a frame is $3 \times 1/3$ ms, or 1 ms. *$T = T_{\text{after}}$ before*
 - ④ output transmission link
transmission link bit rate = 3 kbps
 - ⑤ frame rate = 1 k frames per sec

Example 6.6

- Figure 6.14 shows synchronous TDM with a data stream for each input and one data stream for the output.
- The unit of data is 1 bit.
- Find (a) the input bit duration, (b) the output bit duration, (c) the output bit rate, and (d) the output frame rate.

Solution

- The input bit duration is the inverse of the bit rate: $1/1 \text{ Mbps} = 1 \mu\text{s}$.
- The output bit duration is one-fourth of the input bit duration, or $\frac{1}{4} \mu\text{s}$. *output bit duration.*
- The output bit rate is the inverse of the output bit duration or $1/(\frac{1}{4} \mu\text{s})$ or 4 Mbps. This can also be deduced from the fact that the output rate is 4 times as fast as any input rate; so the output rate $= 4 \times 1 \text{ Mbps} = 4 \text{ Mbps}$. *just on synchronous.*
- The frame rate is always the same as any input rate. So the frame rate is 1,000,000 frames per second. Because we are sending 4 bits in each frame, we can verify the result of the previous question by multiplying the frame rate by the number of bits per frame.

$$\text{a) Input time slot} = \frac{1}{1\text{M}} = 1 \mu\text{sec}$$

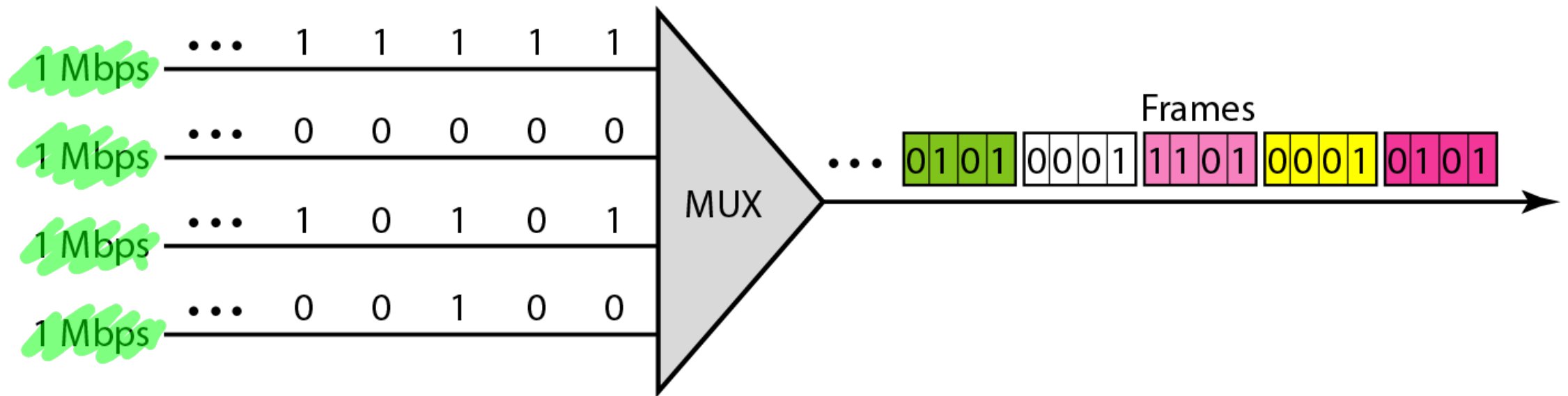
$$\text{c) Frame time} = 1 \mu\text{sec}$$

$$\text{d) output bit rate} = 4 \text{ Mbps}$$

$$\text{b) output bit duration} = \frac{1}{4} \mu\text{s}$$

$$\text{e) Frame rate} = \frac{1}{\text{Frame time}} = \frac{1}{1 \mu\text{s}} = 1 \text{M frame per sec}$$

Figure 6.14 *Example 6.6*



Example 6.7

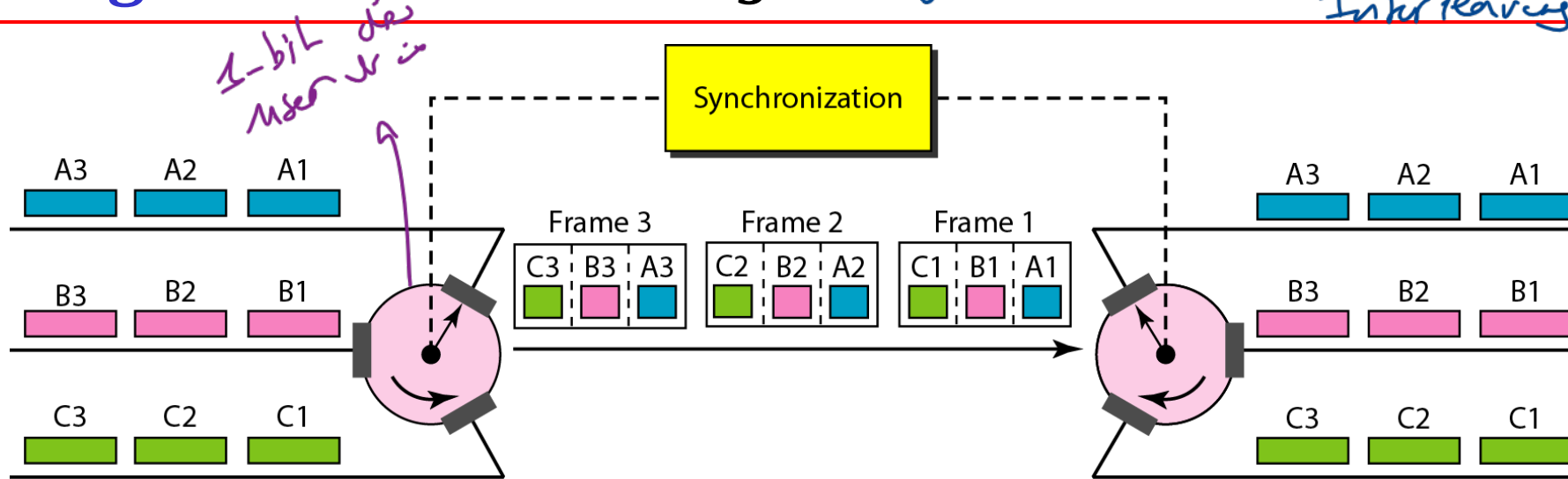
- Four 1-kbps connections are multiplexed together. A unit is 1 bit. *output bit rate*
- Find (a) *Input slot* the duration of 1 bit before multiplexing, (b) the transmission rate of the link, (c) the duration of a time slot, and (d) the duration of a frame. *output*

Solution

- The duration of 1 bit before multiplexing is $1 / 1 \text{ kbps}$, or 0.001 s (1 ms).
- The rate of the link is 4 times the rate of a connection, or 4 kbps.
- The duration of each time slot is one-fourth of the duration of each bit before multiplexing, or $1/4 \text{ ms}$ or $250 \mu\text{s}$.
 - Note that we can also calculate this from the data rate of the link, 4 kbps. The bit duration is the inverse of the data rate, or $1/4 \text{ kbps}$ or $250 \mu\text{s}$.
- The duration of a frame is always the same as the duration of a unit before multiplexing, or 1 ms . We can also calculate this in another way. Each frame in this case has four time slots. So the duration of a frame is 4 times $250 \mu\text{s}$, or 1 ms .

$$4 \times 250 \mu\text{s}$$

Figure 6.15 Interleaving :- single bit to create a frame called Interleaving.



- TDM can be visualized as two synchronized fast-rotating switches, one for muxing and the other for demuxing
- The process of inserting one unit of data per connection in each frame is called Interleaving and the data unit is called the interleaved unit

Example 6.8

- Four channels are multiplexed using TDM.
- If each channel sends 100 bytes/s and we multiplex 1 byte per channel,
- ① show the frame traveling on the link, the size of the frame, the duration of a frame, the frame rate, and the bit rate for the link.

Input data unit.

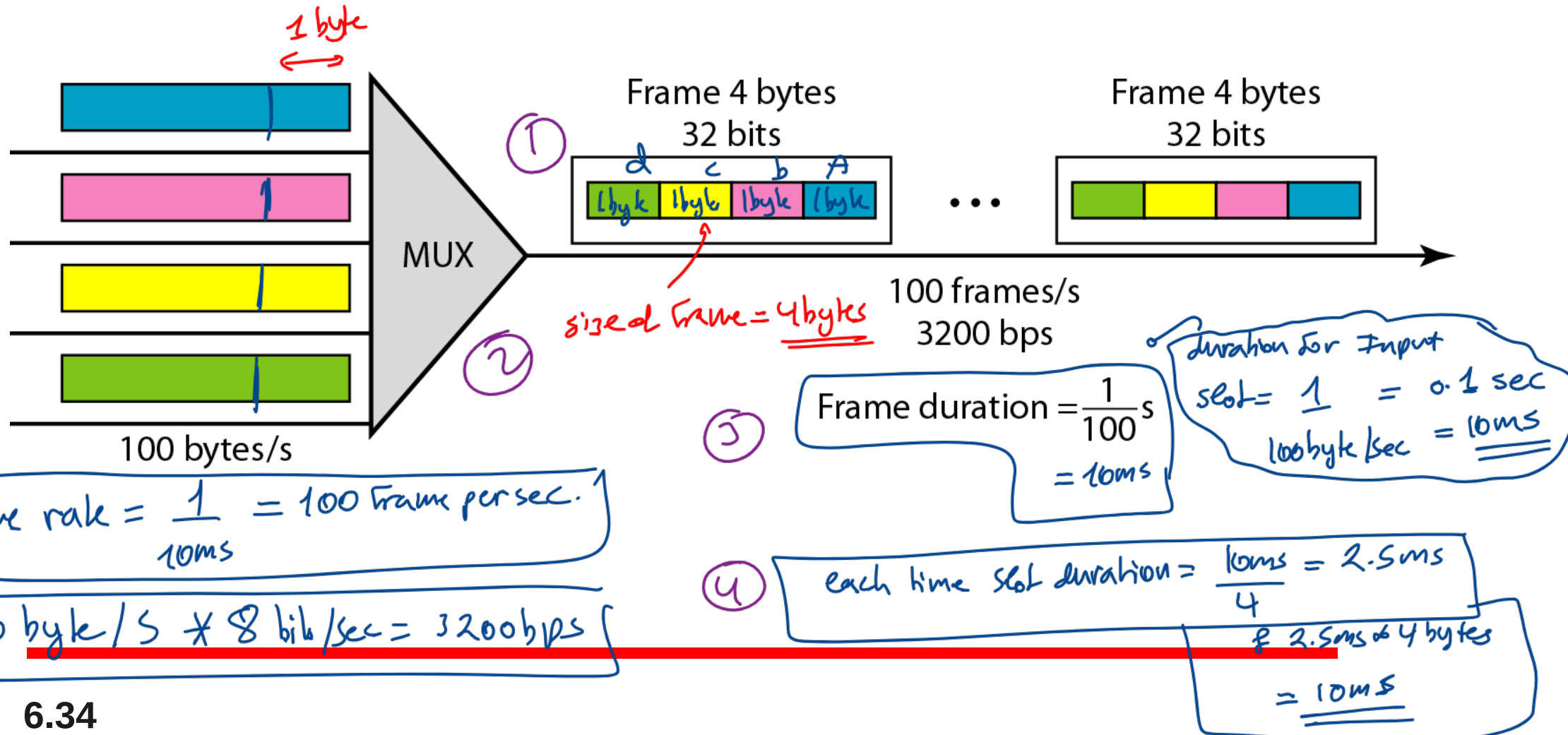
④ each slot duration

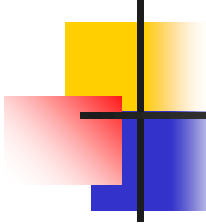
Solution

- The multiplexer is shown in Figure 6.16.
- Each frame carries 1 byte from each channel;
- The size of each frame, therefore, is 4 bytes, or 32 bits.
- Because each channel is sending 100 bytes/s and a frame carries 1 byte from each channel, the frame rate must be 100 frames per second.
- The bit rate is 100×32 , or 3200 bps.

{data rate not bit rate}

Figure 6.16 Example 6.8





Example 6.9

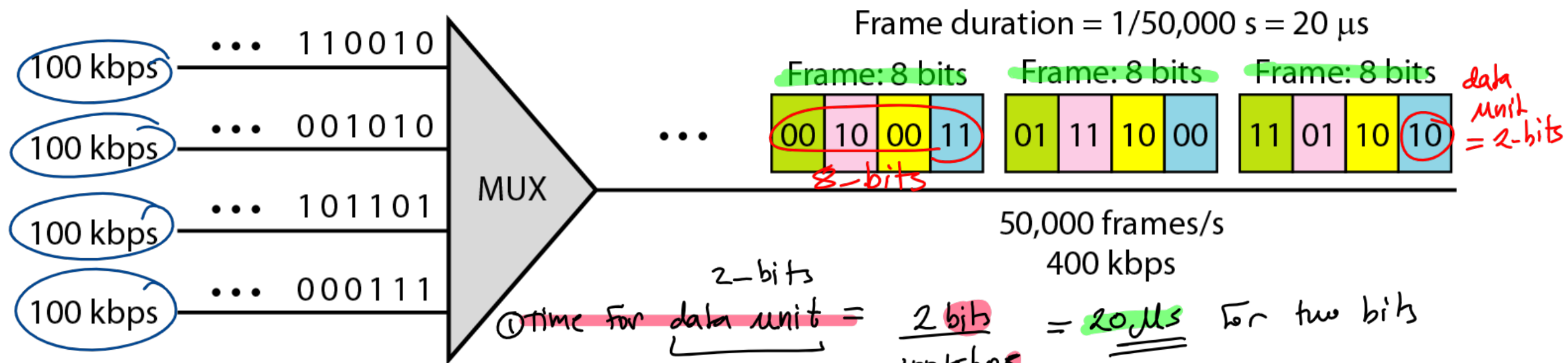
- A multiplexer combines four 100-kbps channels using a time slot of 2 bits. ^{data units} Show the output with four arbitrary inputs.
 - What is the frame rate?
 - What is the frame duration?
 - What is the **bit** rate?
 - What is the **bit** duration?

حسب دو طالب ا.

Solution

- Figure 6.17 shows the output for four arbitrary inputs.
 - The link carries 50,000 frames per second.
 - The frame duration is therefore $1/50,000$ s or $20 \mu\text{s}$.
 - The frame rate is 50,000 frames per second, and each frame carries 8 bits;
 - The bit rate is $50,000 \times 8 = 400,000$ bits or 400 kbps.
 - The bit duration is $1/400,000$ s, or $2.5 \mu\text{s}$.

Figure 6.17 Example 6.9



③ Time for frame = $4 \times 5 \mu$
 Frame duration = $20 \mu\text{sec}$

④ Frame rate = $\frac{1}{20 \mu} = 50 \text{ kFPS}$

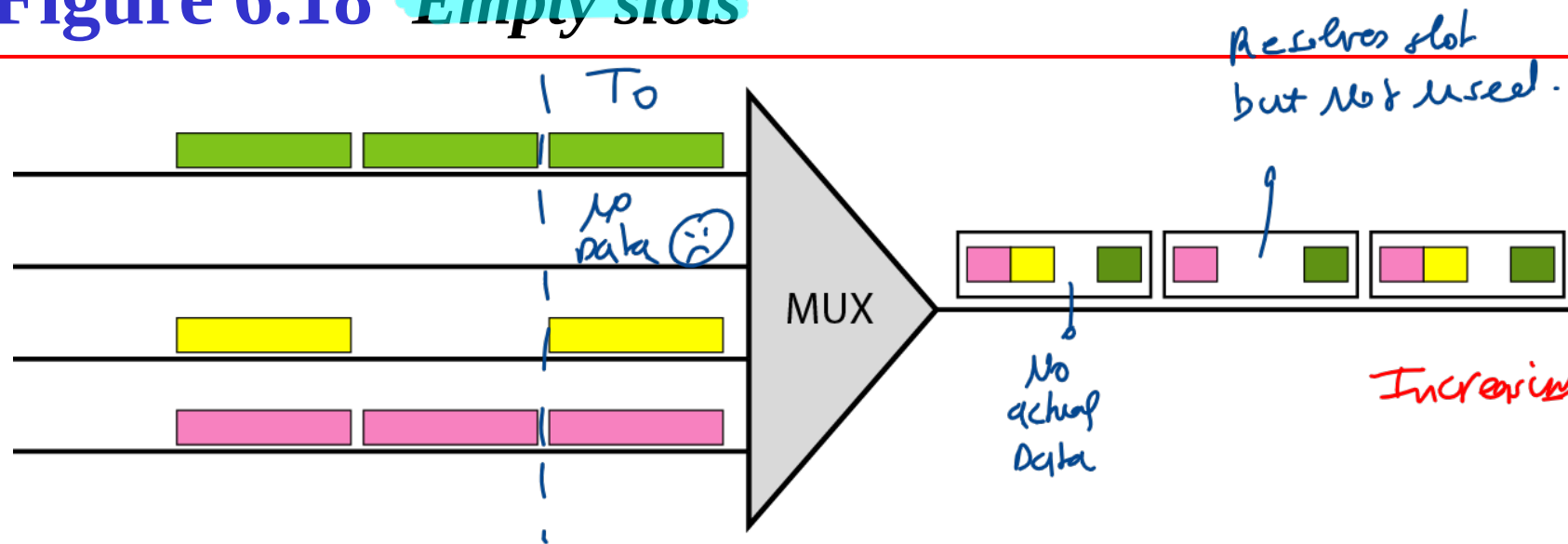
⑤ bit rate = $4 \times 100 \text{ kbps} = 400 \text{ kbps}$
 6.36

⑥ output rate = 400 kbps

② output bit duration = $\frac{5 \mu\text{s}}{2} = 2.5 \mu\text{s}$

problems:-

①
Figure 6.18 Empty slots



Increasing efficiency :-
الوقت لا تضيع .

- Synchronous TDM may not be very efficient
- If no data is available at the input source, then the corresponding time slot in the output frame will be empty → wasting resources
- Statistical TDM can improve the efficiency by removing the empty slots

بجواز Time slot لا يترك .

② Non-Homogeneous TDM

{ have different bit rates to send }

↔ user و man کے لیے

- ↳ different users with different data units.

- Can we multiplex devices that have **different** data rates?
- The answer is yes. **How?**
 - **Faster devices** may use **more time slots** per frame than slower devices
 - But remember that in a TDM system:
 - ~~The number of multiplexed devices, the number of time slots per frame, and the time slot duration in the frame are always fixed~~
 - Therefore, for TDM to work **optimally**, the different data rates must be **integer multiples** of each other. In this case, each device uses a number of time slots that corresponds to the ratio of its data rate to data rate of the slowest device
 - For example: for three devices with data rates x , $2x$, and $6x$, the devices are assigned 1, 2, and 6 time slots per frame respectively.
 - What if data rates are not integer multiples of each others? Use **Bit Padding (or Pulse Stuffing)**
 - Bit Padding is a technique used by the multiplexer to force the data rates of the devices to be integer multiples of each other by adding extra dummy bits in the source streams of the slow device

move slots $\rightarrow$ video bit rate $\rightarrow$ video

- For example: for three devices with data rates x , $2x$, and $6x$, the devices are assigned 1, 2, and 6 time slots per frame respectively.

- What if data rates are not integer multiples of each others? Use **Bit Padding (or Pulse Stuffing)** $\rightarrow$ $\bar{n} \neq m$

- Bit Padding is a technique used by the multiplexer to force the data rates of the devices to be integer multiples of each other by adding extra dummy bits in the source streams of the slow device

20 20 17
padding

the **Bit Padding (or Pulse Stuffing)** $\rightarrow$ کایہ جراتہ
 the data rates of the devices to be integer متساویہ
 source streams of the slow device ہم اضافہ کرتے ہیں
مثلاً اگر ایک ڈیٹا اسٹریم 1000 بت فی سیکنڈ ہے اور دوسرا 1500 بت فی سیکنڈ ہے تو ہم 1000 بت پر 500 بت اضافہ کریں گے تاکہ دونوں ڈیٹا اسٹریمز 1500 بت فی سیکنڈ بن جائیں۔

Figure 6.19 *Multiplexing on two levels.*
Multilevel multiplexing

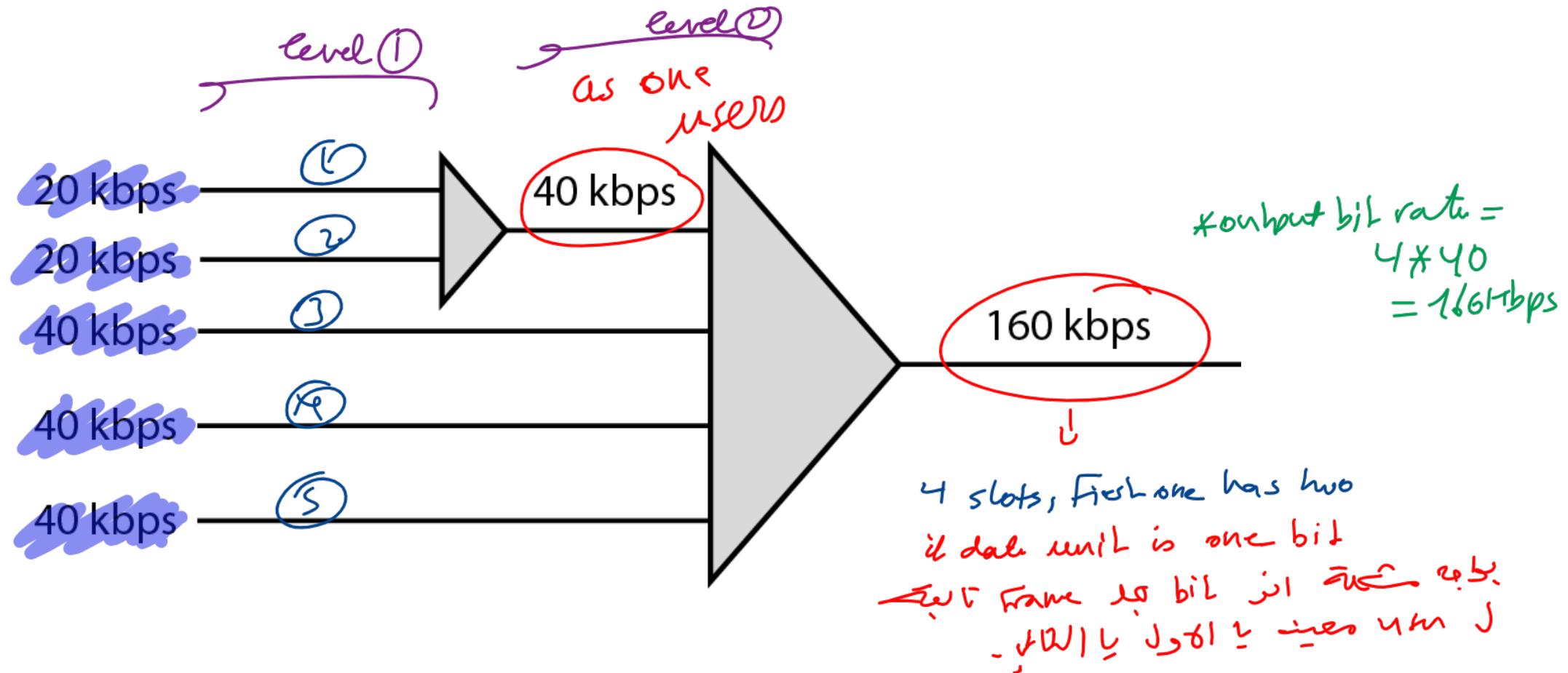


Figure 6.20 *Multiple-slot multiplexing*

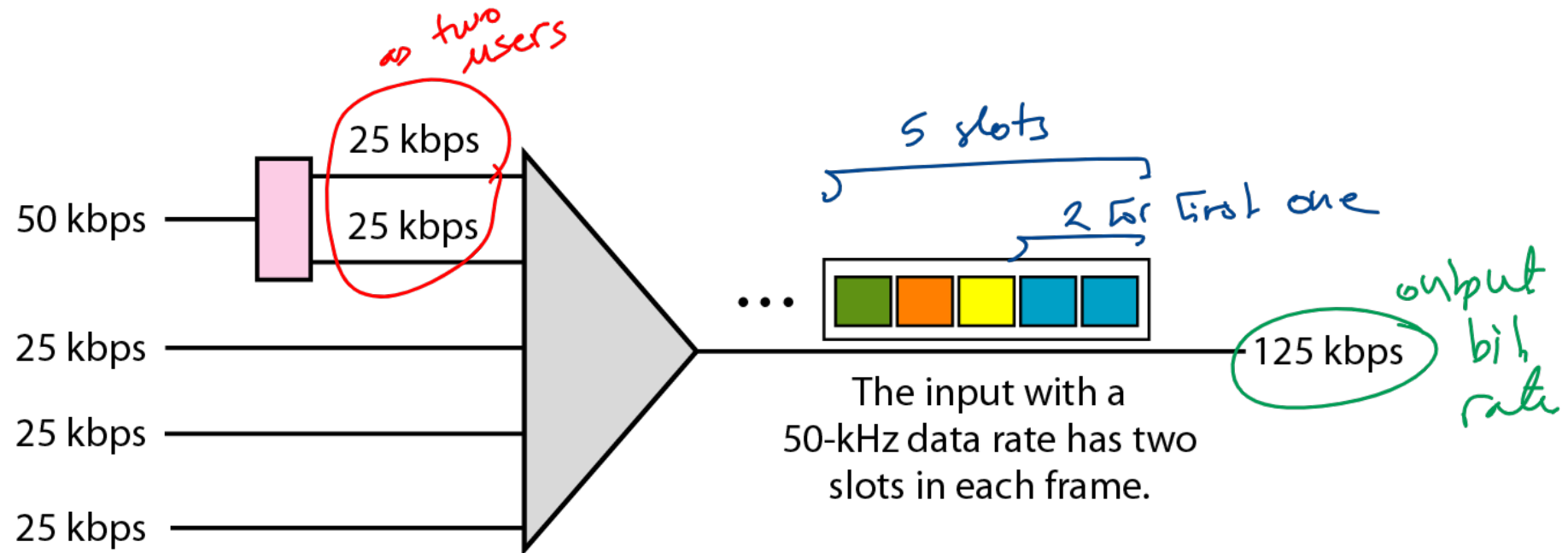


Figure 6.21 *Pulse stuffing*

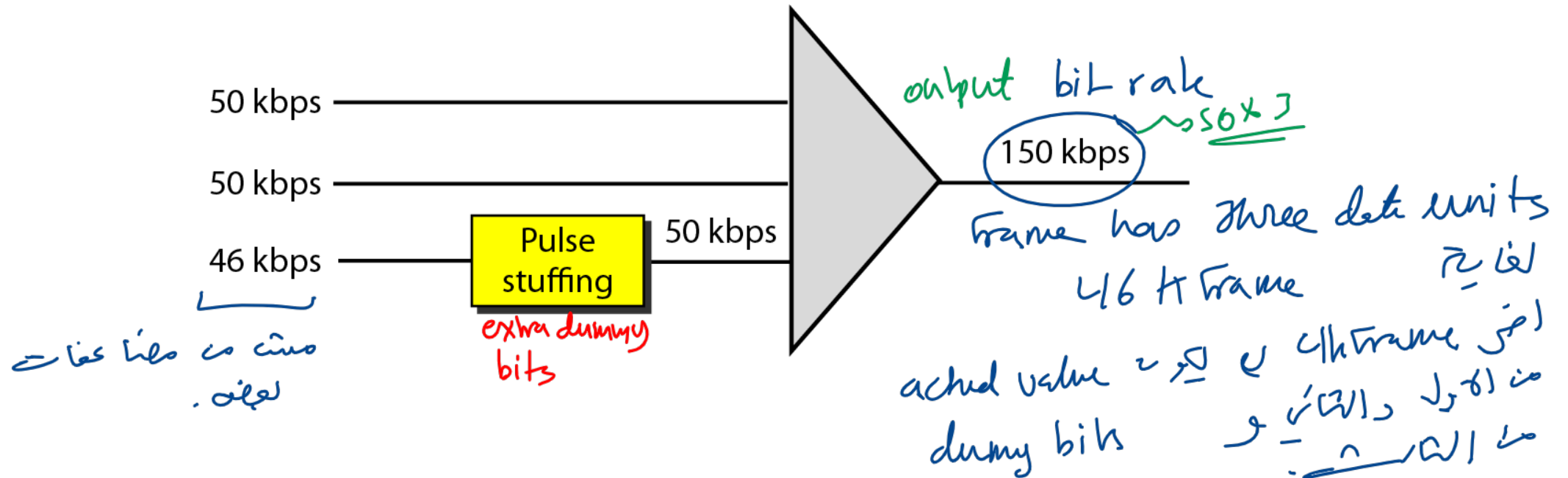
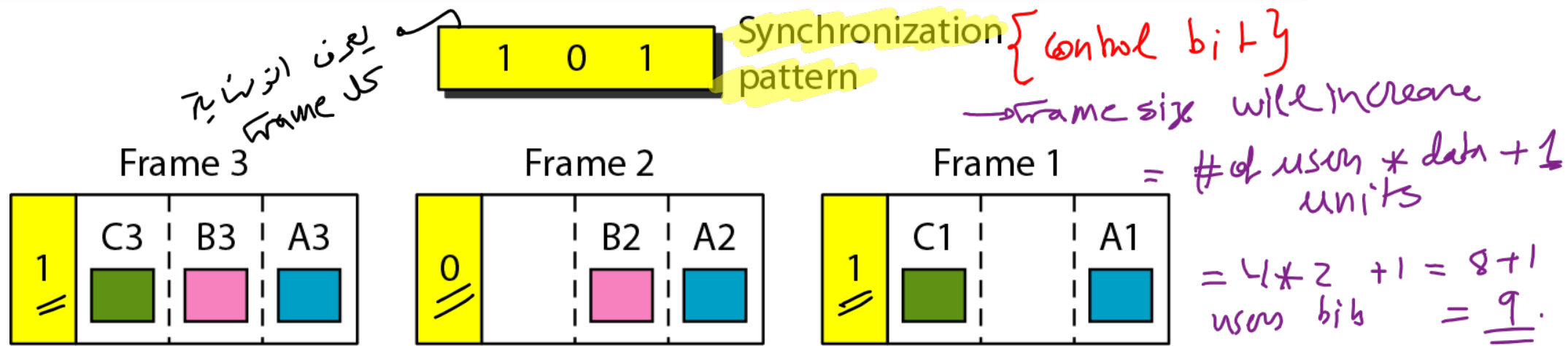


Figure 6.22 Framing bits [adding extra synchronization bits on frame]



- The major issue with TDM is the **synchronization** between the multiplexer and the demultiplexer
- The lack of synchronization may cause the data to be delivered to the wrong channel
- Therefore, one or more synchronization bits (called **framing bits**) are usually added to the beginning of each frame
- The framing bits usually follow a predefined pattern known by both sides such as alternating 1's and 0's

Example 6.10

$$250 \times 8 = 2000 \text{ bps} = 2 \text{ kbps}$$

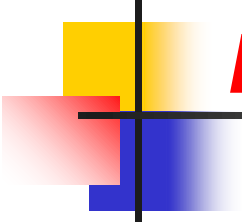
- We have four sources, each creating 250 characters per second.
- If the interleaved unit is a character and 1 synchronizing bit is added to each frame, find
- (a) the data rate of each source, (b) the duration of each character in each source, (c) the frame rate, (d) the duration of each frame, (e) the number of bits in each frame, and (f) the data rate of the link.

Solution

- We can answer the questions as follows:
 - The data rate of each source is $250 \times 8 = 2000 \text{ bps} = 2 \text{ kbps}$.
 - Each source sends 250 ch/sec; therefore, the duration of a character is $1/250 \text{ s}$, or 4 ms.
 - Each frame has one character from each source, which means the link needs to send 250 frames/sec to keep the transmission rate of each source.
 - The duration of each frame is $1/250 \text{ s}$, or 4 ms. Note that the duration of each frame is the same as the duration of each character of each source.
 - Each frame carries 4 characters and 1 extra synchronizing bit. This means that each frame is $4 \times 8 + 1 = 33 \text{ bits}$.
 - The link sends 250 frames per second and each frame contains 33 bits. This means that the data rate of the link is $250 \times 33 = 8250 \text{ bps}$.

$$\begin{aligned} \text{* Frame size} &= 4 \times 8 + 1 \\ &= 33 \text{ bit} \\ \text{* bit duration} &= \frac{8}{2 \text{ kbps}} \end{aligned}$$

$$\begin{aligned} &4 \times 250 \times 8 + \left\{ \begin{array}{l} \text{\# of frames} \\ \text{Frame Rate} \end{array} \right\} \\ &= 4 \times 250 \times 8 + 250 = 8250 \text{ bps} \end{aligned}$$



Example 6.10 (continued)

- Note that the bit rate of the link is ^{8 250} greater than the combined bit rates of the four channels. ⁸⁰⁰⁰
- If we add the bit rates of the four channels, we get 8000 bps.
- Because 250 frames are traveling per second and each contains 1 extra bit for synchronization, we need to add 250 to the sum to get 8250 bps.

Example 6.11

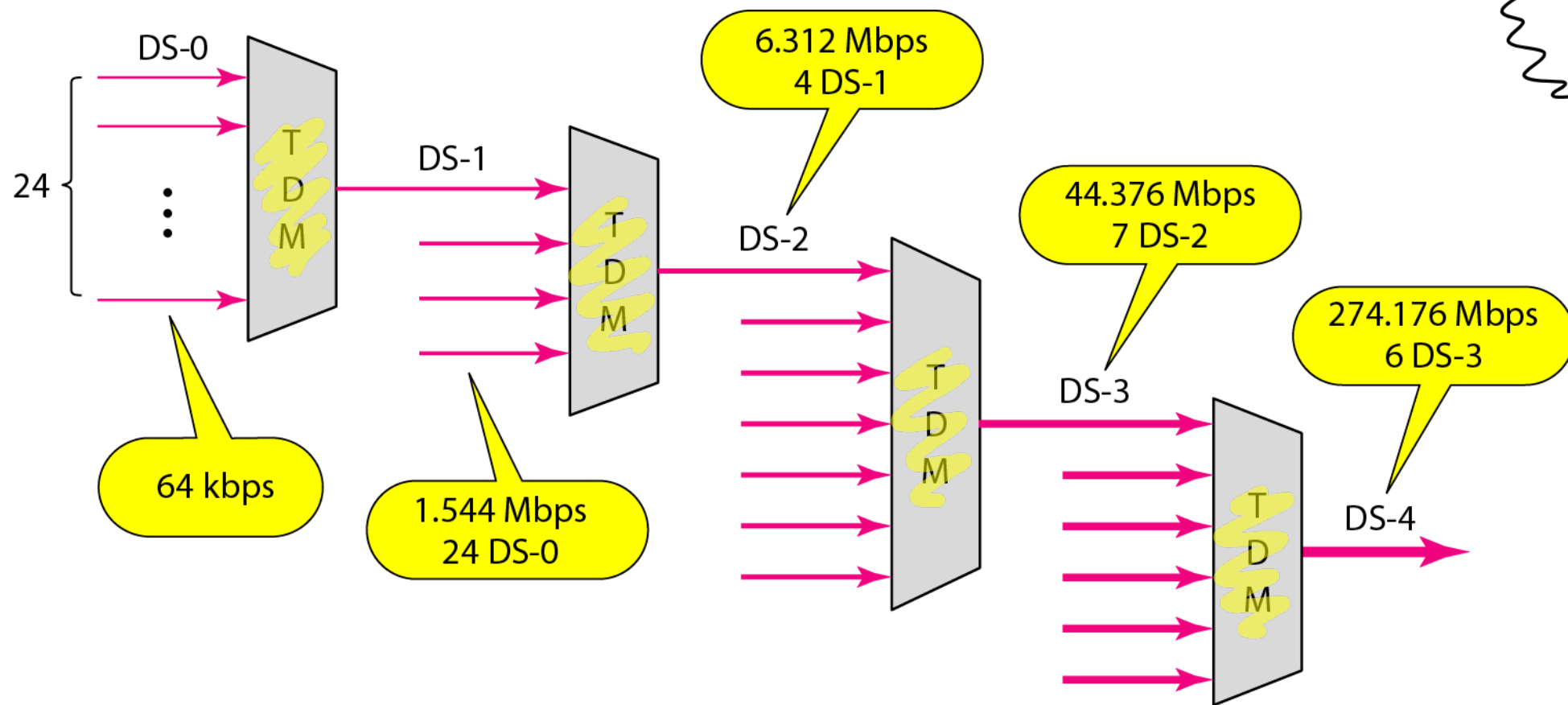
- Two channels, one with a bit rate of 100 kbps and another with a bit rate of 200 kbps, are to be multiplexed. [Non homogeneous synchronous TDM multi-level or mult slots?]
- How can this be achieved? What is the frame rate? What is the frame duration? What is the bit rate of the link? $\leftarrow \text{also } \sim \text{is } 100$
 $\leftarrow \text{also } \sim 81$
 $\leftarrow \text{also } \sim 101 \text{ slots}$

Solution

- We can allocate one slot to the first channel and two slots to the second channel.
- Each frame carries 3 bits.
- The frame rate is 100,000 frames per second because it carries 1 bit from the first channel. * frame duration = $\frac{1}{100k} = 10 \mu s$.
- The bit rate is 100,000 frames/s $\times$ 3 bits per frame, or 300 kbps.
* frame rate = $\frac{1}{10 \mu s} = 100k \text{ frames per sec.}$

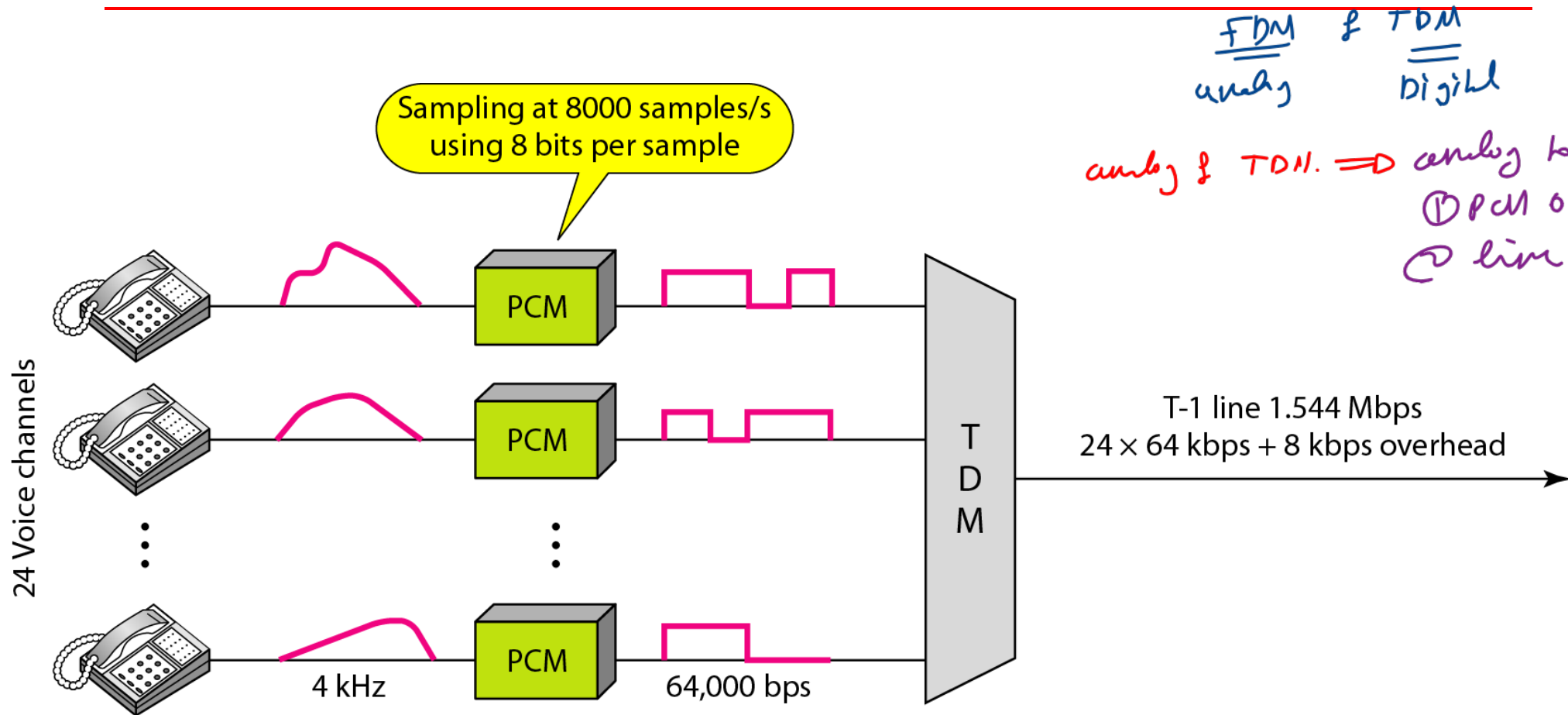


Figure 6.23 Digital hierarchy



*synchronization } data rate 300 kbps + 100 kbps = 400 kbps
 nonsynchronization }
 if synchronization → data rate 300 kbps + 100 kbps = 400 kbps

Figure 6.24 *T-1 line for multiplexing telephone lines*



Synchronous TDM Application

second generation

- Second-generation cellular phone technology called **TDMA (Time-Division Multiple Access)**: *sharing time to access to transmission media.*
 - Digital version of the AMPS technology
 - FDM is still used to divide the bandwidth into 30 KHz bands
 - TDM is used for each FDM band so that 6 phone calls can share the band
 - Therefore, 6 time-slot frames are used for each band
 - That is, the capacity is increased by 6 times over AMPS

790 users in AMPS
تحت 800
لوي

$6 \times 800 = 4800$ users
Transmission media all going
at the same time.

Synchronous TDM $\Rightarrow$ 5 users, frame 5 slots
 $\rightarrow$ 2 users are sending
efficiency $= \frac{2}{5} = 40\%$ bad!

Statistical Time-Division Multiplexing

م
معدل انتقال البيانات
data rate
معدل انتقال البيانات
data rate

- Synchronous TDM is inefficient when there are empty slots

- In statistical TDM: حسب عدد المستخدمين

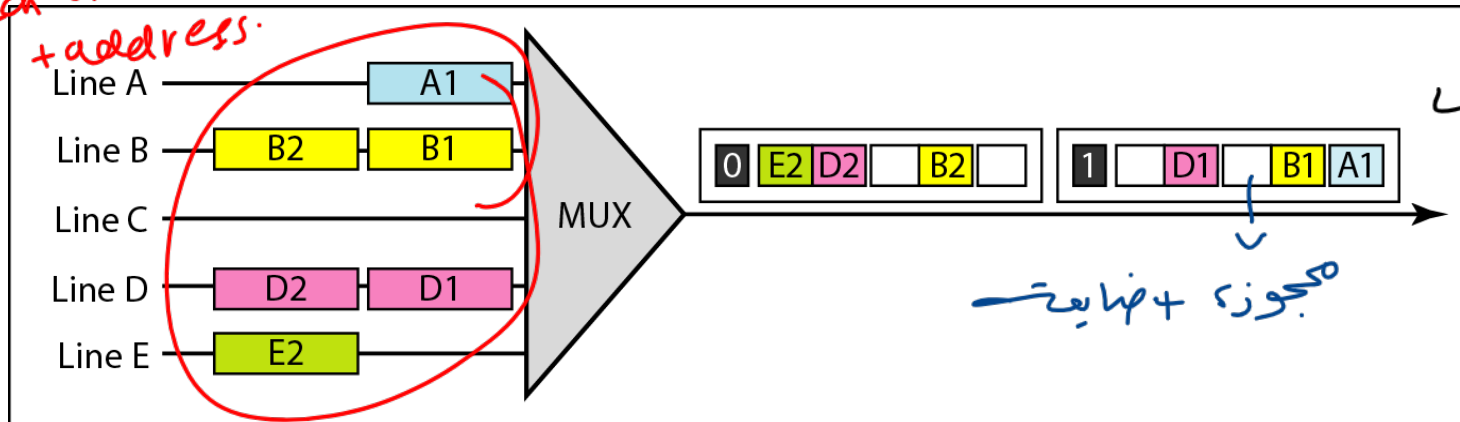
معدل انتقال البيانات
frame

- The time-slots are dynamically allocated to improve bandwidth efficiency or utilization
- The number of slots in each frame is less than the number of input lines
4 users → current want 2 to send,
frame becomes 2 slots.
- Addressing per slot is needed to specify the destination
- The ratio of the data size to the address size must be reasonable
at least 50%
actual data smaller in time slot.
- No frame synchronization bit would be needed
- The bandwidth of the link is usually less than the sum of the bandwidth requirements of all input channels
len frame size.
- The link bandwidth is usually sized based on the traffic intensity and statistics

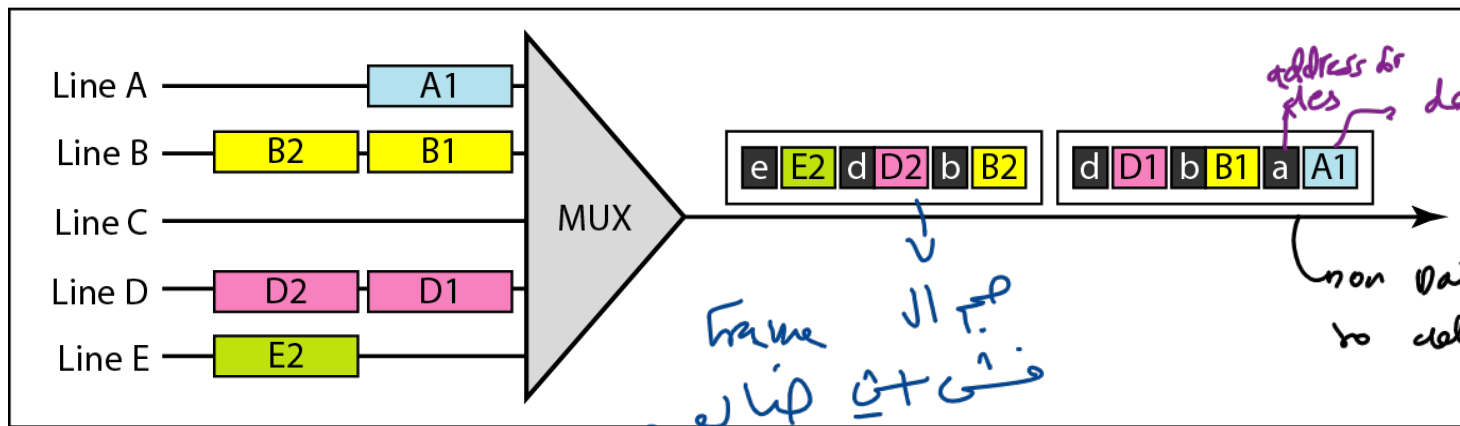
تقليل كفاءة المجرى والفاصلة
adv:-
معيب يزي اننا نوصف
disadv:-
نكسر

Figure 6.26 TDM slot comparison

each one has
data + address.



a. Synchronous TDM



b. Statistical TDM

Questions- 6 channels, 50 kbps

Data unit is 3 bits

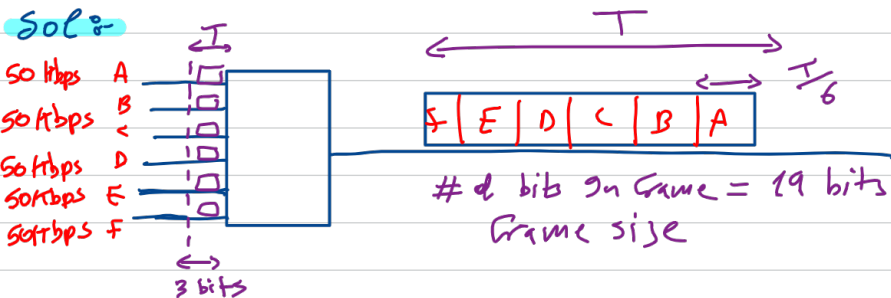
assume we are adding Frame bits to the Frame.

① Draw output frame ② Time for Input slot for Input bit

③ Time for output slot for output bit ④ Frame time ⑤ Frame rate

⑥ output bit rate

Sol:-



① Input slot duration = $\frac{3}{50 \text{ kbps}} = 0.06 \text{ ms} = \text{frame duration}$

② Input bit duration = $\frac{1}{50 \text{ k}} \text{ or } \frac{0.06 \text{ ms}}{3} = 0.02 \text{ ms}$

③ output slot duration = $\frac{0.06}{6} = 0.01 \text{ ms}$

④ output bit duration = $\frac{0.06 \text{ ms}}{19} \text{ or } \frac{0.01 \text{ ms}}{3} = 0.0033 \text{ ms}$

⑤ Frame duration = $\frac{3}{50 \text{ kbps}} = 0.06 \text{ ms}$

⑥ Frame rate = $\frac{50 \text{ kbps}}{3} = 16.67 \text{ Kfps}$

⑦ output bit rate = $\sum N_{\text{input}} + \text{frame rate} = 316.67 \text{ kbps}$

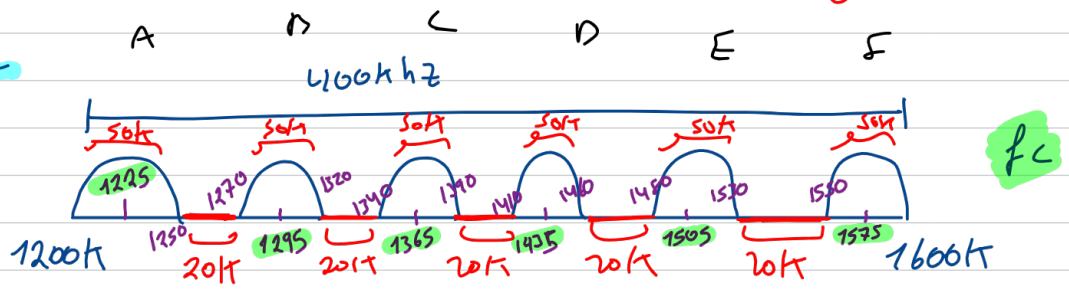
.....

Question:- 6 channels, 50KHz
 Available BW = 400KHz
 $f_{min} = 1.2 \text{ MHz}$

[show FDM using the full available BW.]
 & what is the carrier frequency for each user

لا فويز 2 توفيز 2
 توفيز 2 زيا 2
 6 زيا 2 BW
 guard bands.

Sol:-



* $50K \times 6 = 300K$
 * $5 \times 18 = 100$
 $gb = 20$

* carrier freq of the C using fifth channel
 * how many guard bands?
 * 9 will leave 10KHz extra guard band becomes 18.

Question:-

6 digital channels, 48Mbps
 BW available Link = 18MHz
 use FDM without guard bands.

Technique content from 48Mbps to 3MHz.

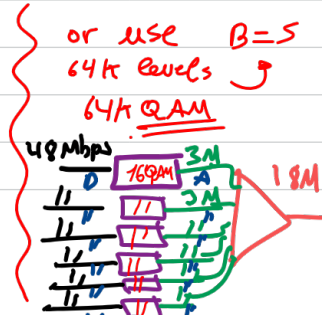
Sol:-

BW for each channel = $\frac{18M}{6} = 3MHz$ {Need to convert} Digital to analog

* Using ASK
 $BW = S$
 $3M = S$
 $S = \frac{N}{r}$
 $3M = \frac{48M}{r}$
 $r = 16 \text{ bits}$

* Using PSK
 same as ASK
 16 bits

* Using FSK
 $BW = 2 \times S$
 $3M = 2 \times S$
 $S = \frac{N}{r}$
 Not enough info



ASK using 16 bits = 64K ASK hard. Too large

6-1 SPREAD SPECTRUM to achieve privacy & security

1 MHz $\rightarrow$ 4 MHz or user 1 & share between multiple users is multiplexing 1/4, BW 1/4
4 MHz $\rightarrow$ 1 MHz is FDM 1/4
TDM 1/4

- In spread spectrum (SS), we combine signals from different sources to fit into a larger bandwidth, but our goals are to prevent eavesdropping and jamming. 1 MHz $\rightarrow$ 4 MHz spread sp. 1/4, $\rightarrow$ 1 MHz $\rightarrow$ 4 MHz as long as have data to send 1/4 or 1 MHz $\rightarrow$ 4 MHz on different carrier freq. Data 1/4 or 1/4
- To achieve these goals, spread spectrum techniques add redundancy. hacker & 1/4

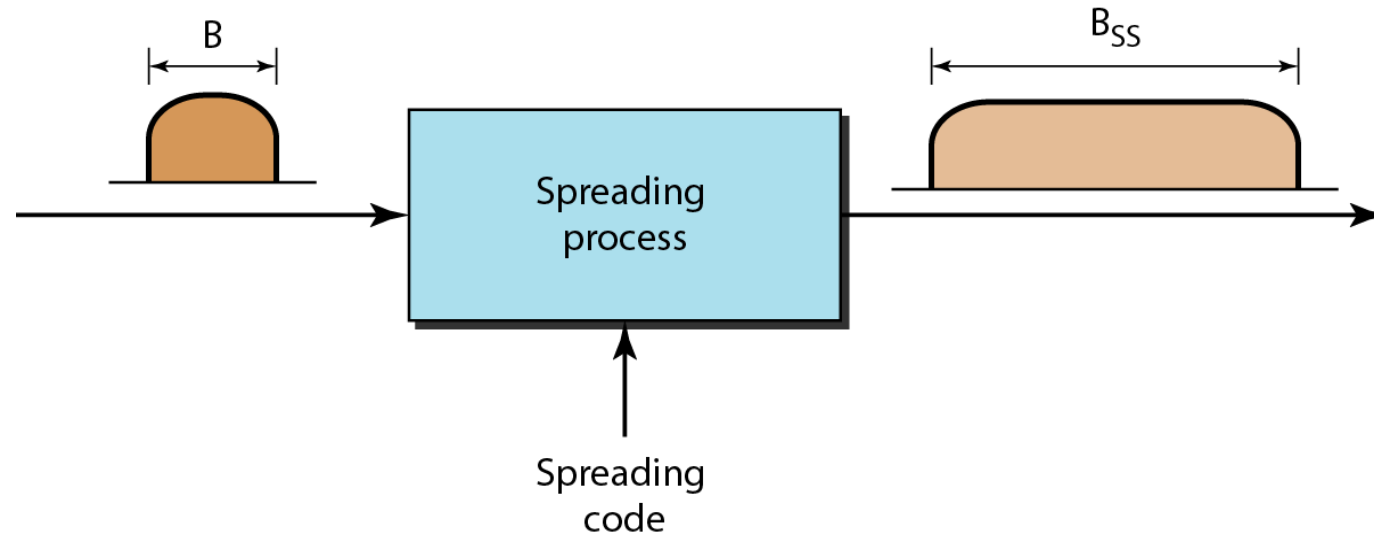
Topics discussed in this section:

- Frequency Hopping Spread Spectrum (FHSS)
- Direct Sequence Spread Spectrum (DSSS)

Listening 1/4 hacker & carrier freq 1/4
Data 1/4 $\rightarrow$ 1/4

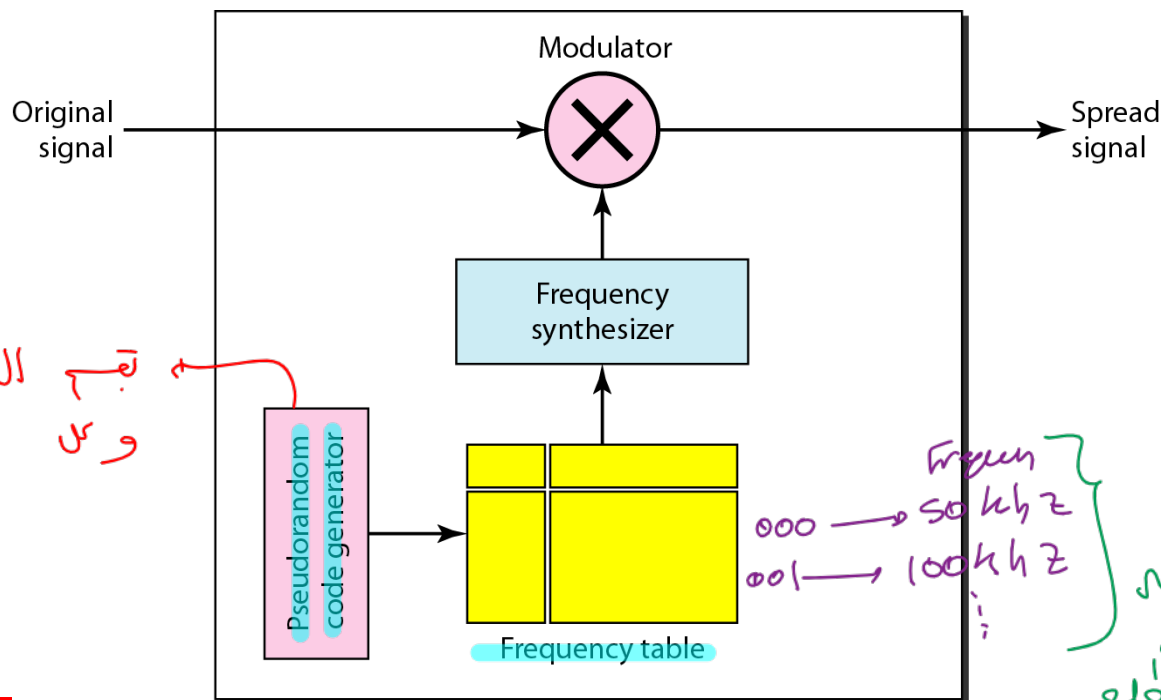
Figure 6.27 Spread spectrum

The whole bandwidth is used by one user each time on different carrier frequency.



- Principles of spread spectrum:
 1. The bandwidth allocated is much larger than needed allowing redundancy
 2. The spreading process occurs after (i.e.; independent of) the signal creation
- The two most well-known spreading techniques used are:
 1. Frequency Hopping Spread Spectrum (FHSS)
 2. Direct Sequence Spread Spectrum (DSSS)

Figure 6.28 Frequency hopping spread spectrum (FHSS) ①



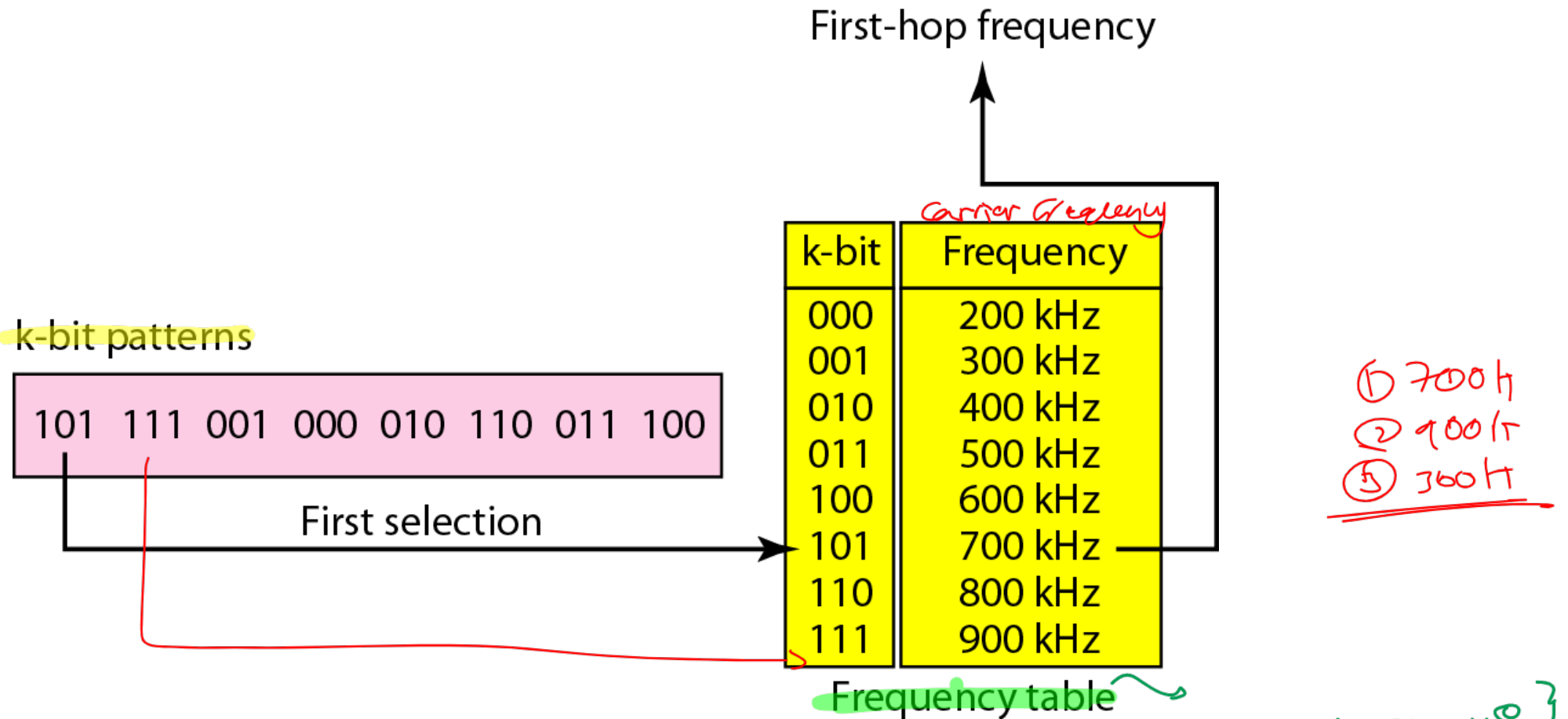
groups of bits
different carrier freq.

data belongs to the same user
different carrier frequencies!
Frequency = 50 kHz

data belongs to the same user
different carrier frequencies!

- M different carrier frequencies to modulate the signal
- A pseudorandom code generator, also called pseudorandom noise (PN) creates a k-bit pattern for every hopping period. *frequency hopping*
- A frequency synthesizer creates a carrier signal with the corresponding frequency according to a lookup table. *different carrier frequency*
- Advantages: privacy and antijamming

Figure 6.29 *Frequency selection in FHSS*



{ Connection establish is sending al Receiver وال sender ال }
وال hacker متي لازم يعرف

Figure 6.30 *FHSS cycles*

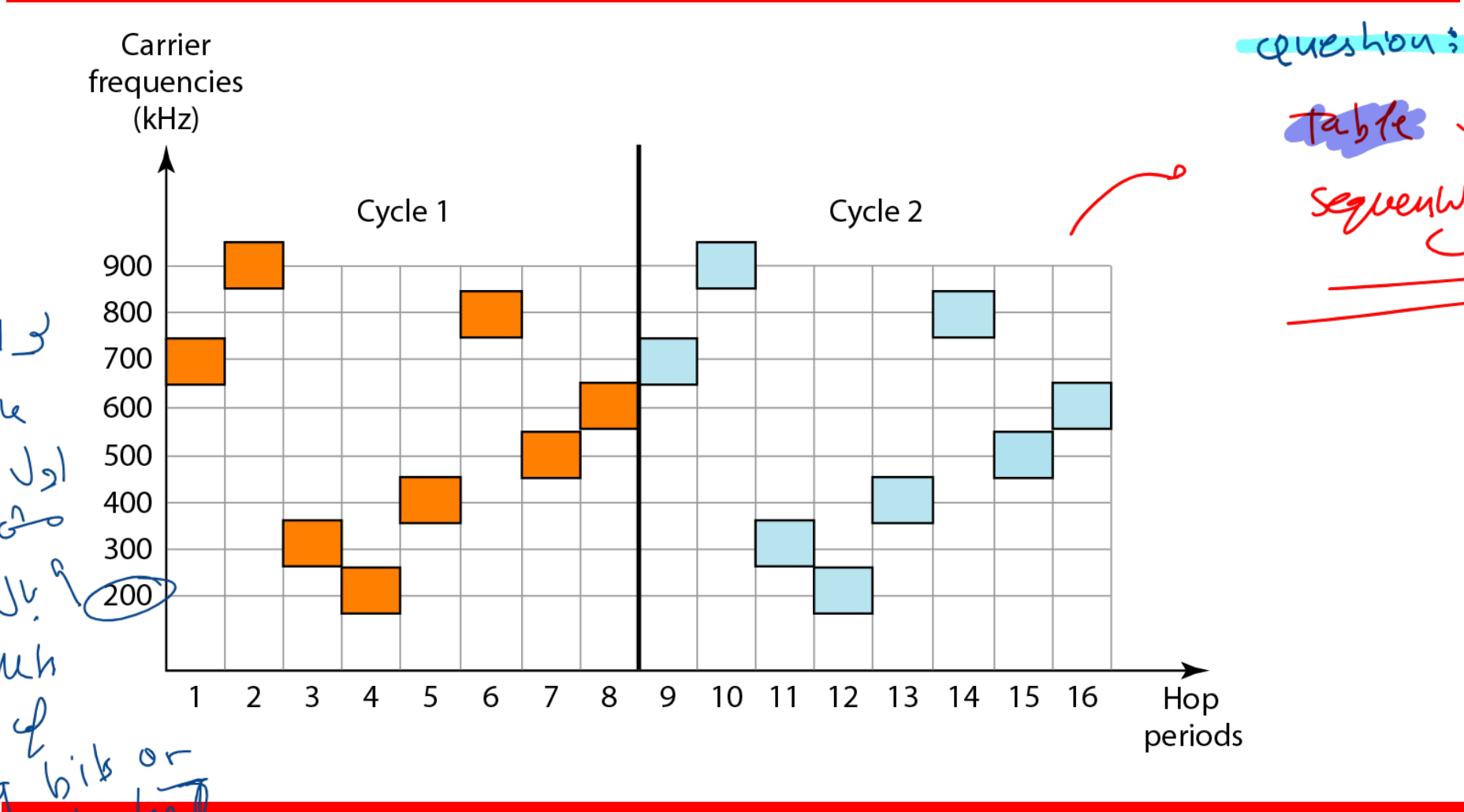
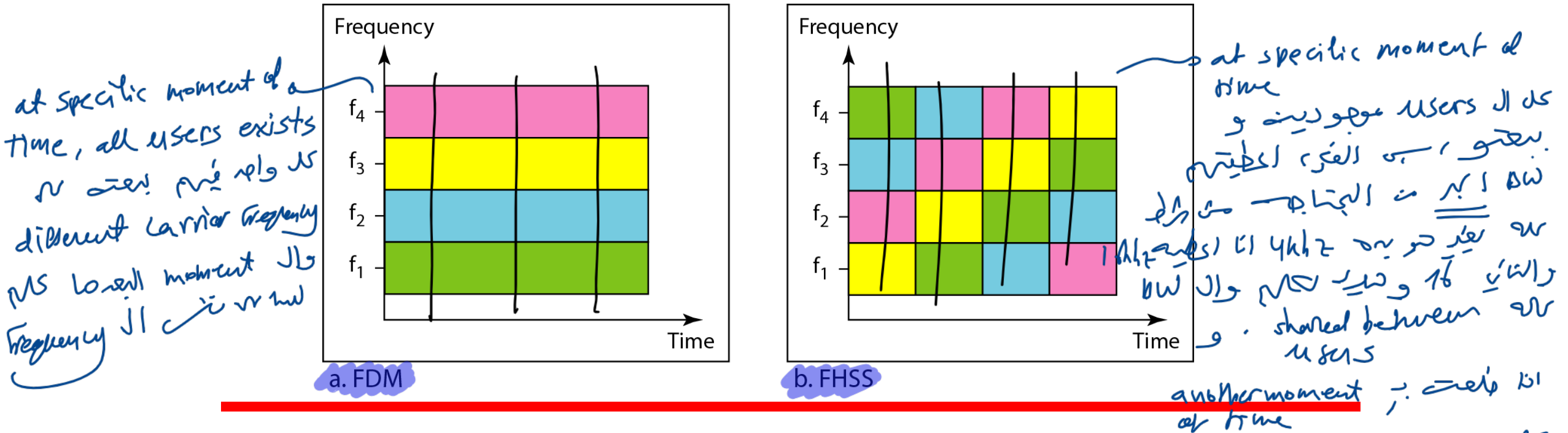


Table 11 + sequence

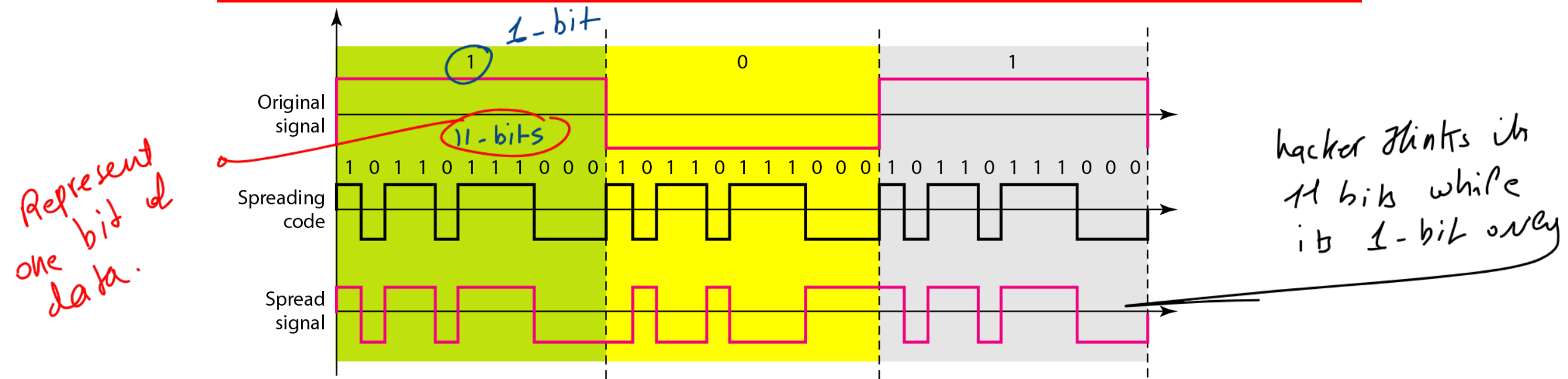
6.58

Figure 6.31 *Bandwidth sharing*



- For M hopping frequencies, M channels can be multiplexed into 1
- In FDM, each channel uses $1/M$ of the bandwidth with fixed allocation
- In FHSS, each channel uses $1/M$ of the bandwidth but with a dynamic allocation from hop to hop

Figure 6.33 DSSS example



- Barker Sequence with $n=11$ is used in wireless LAN [1-bit $\rightarrow$ 11-bit]
- Assume that the original signal and the chips use polar NRZ encoding $\rightarrow$ DC component.
- The rate of the spreading signal is 11 times that of the original signal. Therefore, the required bandwidth of the spreading signal is 11 times that of the original
- Advantages: privacy and immunity to interference
- Can we share the bandwidth in DSSS?
 - Depends on the orthogonality of the spreading code
 - If different channel is assigned a different code from a set of orthogonal codes, then yes.

1 bit $\xrightarrow{\text{noise}}$ 100% error

11-bit $\xrightarrow{\text{noise}}$ 10% error

actual data